Dyadic-Comb Frontiers for the Fabius–Rvachev System

Comb sums, exact Euler–Maclaurin collapse, spectral Dirichlet values, shifted quadrature, and global polynomial interpolation on the dyadic comb

Consolidated frontier volume for Vladimir Reshetnikov's `ProveIt` project

28 August 2026

Abstract

Fix a dyadic level m, put $M = 2^m$ and $h = 1/M$, and sample the Fabius function F or Rvachev's compactly supported function up on the additive dyadic comb $\{k/M\}$. This volume consolidates six independently written research reports on what those samples can and cannot do, in two parts.

Part I: sums and quadrature. The weighted comb sums $h\sum_k (k/M)^\alpha F(k/M)$ are far more rigid than generic Riemann sums. The complete sample row has a rational generating function whose apparent pole of order $m+1$ collapses to a single simple pole; Prouhet cancellation compresses the 2^m-term Thue–Morse block of any fixed-degree sum to at most $p+2$ Bell-polynomial terms; and the Euler–Maclaurin expansion of the degree-p comb is not merely asymptotic but *exactly* terminating as soon as $m \geq \tau(p) := 2\lceil p/2 \rceil$. The engine is the order-$m+\nu_2(\ell)$ zero of the Fabius characteristic function at the dual lattice. The first failure is a Thue–Morse–signed half-integer sinc-product Dirichlet series $D(s) = \sum_{q \text{ odd}} \widehat{\mathrm{up}}(q/2)\, q^{-s}$, and the defect identity forces $D(2r)/\pi^{2r} \in \mathbb{Q}$ with

$$D(2) = \frac{\pi^2}{18}, \qquad D(4) = \frac{23\pi^4}{4050}, \qquad D(6) = \frac{1543\,\pi^6}{2679075}, \qquad D(8) = \frac{1196113\,\pi^8}{20494923750}.$$

For the full Rvachev comb the quadrature theorem is uniform in an arbitrary real shift: every translate of the level-m comb integrates every polynomial of degree at most m exactly, with an explicit finite 2-adic alias hierarchy, phase superconvergence at the first failing degree, and a complete Bell–Bernoulli calculus for all higher alias jets. Complex, fractional, and negative powers are governed by an entire Hurwitz-zeta representation, a universal singular correction $2\zeta(-\alpha)h^{\alpha+1}$, and a logarithmic transition at $\alpha = -1$; iterated and fractional-order summation reduce exactly to shifted one-fold combs.

Part II: global interpolation. Interpolating the same exact rational data by one global polynomial is a sharp laboratory for Runge instability. Ordinary interpolation undergoes a dramatic transition (sampled errors 5, 3.6×10^7, 3.8×10^{22} at $M = 32, 64, 128$), even though the samples are exact and the target is C^∞ and infinitely flat at the endpoints. Imposing zero endpoint jets of order r has the exact smoothstep factorization $H_{M,r} = S_r + [x(1-x)]^{r+1}Q_{M,r}$, which converts the confluent problem into interior barycentric interpolation and yields explicit cardinal modes. Balancing two modes predicts the observed optimal jet order $r \asymp M\log 2/\log M$ almost perfectly, and tuned jets temporarily reach errors near 10^{-6}; but no schedule $r = r(M)$ stabilizes the operator: the weighted Lebesgue constant satisfies $\mathscr{L}_{M,r} \geq C\mathrm{e}^{cM}$ uniformly in r, and by $M = 256$ even the optimized error diverges. Matching derivatives at every node is strictly worse (the central cardinal grows like $4^M/M^4$), while local Hermite interpolation on the same comb is uniformly stable with explicit algebraic bounds. Exact endpoint-derivative defects, a 2-adic valuation law for the barycentric weights, Lambert-W peak migration, and Thue–Morse cancellation phenomena connect the instability to the surrounding arithmetic of the corpus.

All theorems come with complete proofs; numerical claims are exact rational computations or high-precision scans with stated grids; and conjectures are labeled as such. Provenance of the six absorbed reports, with SHA-256 hashes, is recorded in Appendix E.

Contents

Provenance and merge conventions

This volume is an *editorial consolidation* (28 August 2026) of six draft reports that arrived independently in the second and third draft waves, all concerned with the additive dyadic comb. Unlike the mechanical part-by-part consolidations elsewhere in this corpus, the six sources here largely re-derive one another's core results, so their content was merged: shared theorems are stated once in a single notation, the clearest available proof of each result was selected (with alternative arguments kept as remarks when they illuminate a different mechanism), and all source-specific material — exact tables, experiments, conjectures, and research programs — was retained. Section E lists the absorbed documents with SHA-256 hashes of their sources and records what was deduplicated; their supporting data, figures, and scripts live under `assets/`. A small number of exact values new to this volume (the spectral Dirichlet values $D(6)$ and $D(8)$ in section 6) were computed for the merge and cross-validated against the absorbed tables; the verification route is recorded where they appear.

Status discipline

A statement presented as a theorem, proposition, lemma, or corollary has a complete proof in this volume (or is explicitly cited to the repository's primary exposition). A *numerical observation* is an exact-rational or high-precision computation with a stated grid, not a continuum claim. *Conjectures*, *problems*, and *research questions* are deliberately separated from the proved results. Nothing here has been formalized in Lean unless the text says so; novelty claims are relative to the audited repository corpus and a targeted literature search, not to all unpublished mathematics.

Unified notation

The six sources used three inconsistent conventions for the comb size ($N = 2^n$, $N = 2^m$, $M = 2^N$). Throughout this volume:

$m \in \mathbb{N}_0$	the dyadic level
$M = 2^m$, $h = 1/M$	the comb size and mesh
$x_k = k/M = kh$	the comb nodes
F	the Fabius function on $[0, 1]$, $F(0) = 0$, $F(1) = 1$
up	Rvachev's even function on $[-1, 1]$, $\mathrm{up}(0) = 1$
$\rho = F'$	the Fabius probability density
$\varepsilon_a = (-1)^{s_2(a)}$	the Thue–Morse sign, s_2 the binary digit sum
$\Theta_m(z) = \prod_{j<m}(1 - z^{2^j})$	the finite Thue–Morse product
$\Phi = \widehat{\mathrm{up}}$	the infinite sinc product (1.8)
c_r, d_s, I_p, $I(\alpha)$	moments; see section 1.3
G_m, C_m	the Appell moment polynomial and dyadic prefactor (1.28)
$L_{m,\alpha}$, $R_{m,\alpha}$, $T_{m,p}$	left, right, trapezoidal Fabius combs (section 2)
$Q_{m,p}(\theta)$, $A_m(\alpha)$	shifted and absolute Rvachev combs (sections 7 and 8)
$\mathcal{I}_M$, $H_{M,r}$, S_r, W_r	interpolation operators of Part II (sections 14 and 15)
$\mathscr{L}_{M,r}$	the weighted Lebesgue constant (16.2)

Falling factorials are written $x^{\underline{k}} = x(x-1)\cdots(x-k+1)$, rising factorials $x^{\overline{k}}$, unsigned Stirling numbers of the first kind $\left[{r \atop k}\right]$, Bernoulli numbers and polynomials B_n and $B_n(x)$, and complete exponential Bell polynomials $Y_r(x_1, \ldots, x_r)$. The Fourier transform convention is $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi\mathrm{i}\xi x}\,\mathrm{d}x$.

1 Common foundations

Everything in this volume rests on a small set of established facts about the Fabius–Rvachev system, all proved in the repository's primary exposition [34] and, where indicated, in its Lean development. This section fixes them in the unified notation; nothing here is claimed as new.

1.1 Normalization and the probability model

Let $U_1, U_2, \ldots$ be independent random variables uniform on $[0,1]$ and put

$$Y = \sum_{j=1}^{\infty} 2^{-j} U_j, \qquad X = 2Y - 1 = \sum_{\nu=1}^{\infty} 2^{-\nu} V_\nu, \tag{1.1}$$

where the $V_\nu = 2U_\nu - 1$ are uniform on $[-1,1]$. The Fabius function is the distribution function of Y,

$$F(x) = \mathbb{P}(Y \le x) \qquad (0 \le x \le 1), \tag{1.2}$$

and Rvachev's function up is the density of X: the unique even C^∞ function supported on $[-1,1]$ with $\int_{-1}^{1} \mathrm{up} = 1$. The two are folds of each other,

$$\mathrm{up}(x) = F(1 - |x|) \quad (|x| \le 1), \qquad \text{so} \qquad \mathrm{up}(x) = 1 - F(x) \quad (0 \le x \le 1), \tag{1.3}$$

and the Fabius density is a rescaled copy of up:

$$\rho(x) := F'(x) = 2\,\mathrm{up}(2x - 1). \tag{1.4}$$

The basic structural identities are the reflection symmetry and the delay-differential self-similarity

$$F(1-x) = 1 - F(x), \qquad F'(x) = 2F(2x) \quad (0 \le x \le \tfrac{1}{2}), \tag{1.5}$$

together with infinite flatness at the endpoints:

$$F^{(q)}(0) = F^{(q)}(1) = 0 \quad (q \ge 1), \qquad \mathrm{up}^{(q)}(\pm 1) = 0, \quad (\mathrm{up} - 1)^{(q)}(0) = 0 \quad (q \ge 0). \tag{1.6}$$

Equivalently $F(x) = \mathcal{O}(x^A)$ as $x \downarrow 0$ for every $A > 0$; this flatness is what makes negative powers harmless in Part I and what makes endpoint derivative data so tempting in Part II.

Iterating (1.5) through the signed extension $\mathcal{F}$ of F (the solution of $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$) gives the derivative tower and the sharp norms

$$\mathcal{F}^{(q)}(x) = 2^{\binom{q+1}{2}} \mathcal{F}(2^q x), \qquad \left\| F^{(q)} \right\|_{\infty,[0,1]} = \left\| \mathrm{up}^{(q)} \right\|_{\infty,[-1,1]} = 2^{\binom{q+1}{2}} = 2^{q(q+1)/2}. \tag{1.7}$$

The tension between (1.6) and (1.7) — flat endpoints, superexponentially growing global derivative norms, nowhere analyticity — is the single most important analytic feature of the system for this volume: it disables every Taylor- or remainder-based convergence argument while enabling exact spectral and arithmetic ones.

1.2 The Fourier product and its integer zeros

With $\widehat{f}(\xi) = \int f(x) \mathrm{e}^{-2\pi \mathrm{i} \xi x}\, \mathrm{d}x$ and $\mathrm{sinc}_\pi z = \sin(\pi z)/(\pi z)$,

$$\Phi(\xi) := \widehat{\mathrm{up}}(\xi) = \prod_{j=0}^{\infty} \mathrm{sinc}_\pi\!\left(\frac{\xi}{2^j}\right), \tag{1.8}$$

an even, real, rapidly decreasing entire function of exponential type. The characteristic function of the Fabius law is

$$\chi(\xi) := \mathbb{E}\, \mathrm{e}^{-2\pi \mathrm{i} \xi Y} = \widehat{\rho}(\xi) = \mathrm{e}^{-\pi \mathrm{i} \xi}\, \Phi(\xi/2). \tag{1.9}$$

At a nonzero integer n, write $n = \sigma 2^v q$ with q odd, $\sigma = \pm 1$, $v = \nu_2(n)$. Exactly the factors $j = 0, \ldots, v$ of (1.8) vanish at n, each simply, so

$$\mathrm{ord}_{\xi = n}\, \Phi = \nu_2(n) + 1, \tag{1.10}$$

and the leading jet at the zero is explicit [34]: with $d = \nu_2(n) + 1$,

$$\Phi^{(d)}(n) = -\frac{d!}{n^d}\,\Phi\Big(\frac{q}{2}\Big)\,. \tag{1.11}$$

Consequently, at the dual lattice of the level-m comb,

$$\operatorname{ord}_{\xi=M\ell}\chi = m + \nu_2(\ell) \qquad (\ell \neq 0), \tag{1.12}$$

a zero order that *grows with the level.* This linear growth of dual-lattice zero multiplicity is the engine of every exactness theorem in Part I; the half-integer values $\Phi(q/2)$, q odd, which are nonzero and satisfy the sign law of theorem 6.1, control every first failure.

1.3 Moments

Three interlocking moment families appear throughout.

Even Rvachev moments. Let

$$c_r := \int_{-1}^{1} x^{2r}\,\mathrm{up}(x)\,\mathrm{d}x = \mathbb{E}X^{2r}, \qquad c_0 = 1; \tag{1.13}$$

odd moments vanish by symmetry. Splitting off V_1 in (1.1) gives the exact rational recurrence

$$(2r+1)\big(2^{2r}-1\big)c_r = \sum_{j=0}^{r-1}\binom{2r+1}{2j}c_j, \tag{1.14}$$

with first values

$$c_1 = \tfrac{1}{9}, \quad c_2 = \tfrac{19}{675}, \quad c_3 = \tfrac{583}{59535}, \quad c_4 = \tfrac{132809}{32531625}, \quad c_5 = \tfrac{46840699}{24405225075}. \tag{1.15}$$

The moment generating function is the infinite product

$$\mathsf{M}(t) := \mathbb{E}\,\mathrm{e}^{tX} = \prod_{\nu=1}^{\infty}\frac{\sinh(t/2^\nu)}{t/2^\nu}, \tag{1.16}$$

whose logarithmic expansion yields the even cumulants

$$\kappa_{2r} = \frac{2^{2r-1}B_{2r}}{r\,(2^{2r}-1)}, \qquad \mathbb{E}X^n = Y_n(0,\kappa_2,0,\kappa_4,\dots), \tag{1.17}$$

one of several places where complete Bell polynomials organize the dyadic structure.

Fabius-law moments. For $s \in \mathbb{C}$ put

$$d_s := \mathbb{E}Y^s = \int_0^1 x^s \rho(x)\,\mathrm{d}x. \tag{1.18}$$

Flatness of ρ at both endpoints makes d_s an *entire* function of s. For integers,

$$d_n = 2^{-n}\sum_{j=0}^{\lfloor n/2\rfloor}\binom{n}{2j}c_j, \qquad d_1 = \tfrac{1}{2}, \quad d_2 = \tfrac{5}{18}, \quad d_3 = \tfrac{1}{6}, \quad d_4 = \tfrac{143}{1350}, \quad d_5 = \tfrac{19}{270}, \tag{1.19}$$

and d_n also satisfies the self-contained recurrence

$$d_0 = 1, \qquad d_n = \frac{1}{2^n-1}\sum_{q=1}^{n}\binom{n}{q}\frac{d_{n-q}}{q+1} \quad (n \geq 1), \tag{1.20}$$

obtained by conditioning on U_1 in (1.1).

Fabius monomial integrals. For $p \in \mathbb{N}_0$, integration by parts gives

$$I_p := \int_0^1 x^p F(x)\,\mathrm{d}x = \frac{1 - d_{p+1}}{p+1} \in \mathbb{Q}, \tag{1.21}$$

and the same formula defines the entire continuation

$$I(\alpha) := \int_0^1 x^\alpha F(x)\,\mathrm{d}x = \frac{1 - d_{\alpha+1}}{\alpha+1} \qquad (\alpha \in \mathbb{C}), \tag{1.22}$$

the singularity at $\alpha = -1$ being removable with

$$I(-1) = -\left.\frac{\mathrm{d}}{\mathrm{d}s} d_s\right|_{s=0} = \mathbb{E}\big[-\log Y\big] = 0.7582400\ldots. \tag{1.23}$$

All three families are tied together by one exponential generating function.

Theorem 1.1 (Moment EGF). *With* M *as in* (1.16),

$$\sum_{p=0}^{\infty} I_p \frac{t^p}{p!} = \frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t \mathsf{M}(t)}{\mathrm{e}^t - 1}, \tag{1.24}$$

and consequently

$$I_p = \frac{1}{p+1}\left[1 - \sum_{j=0}^{p+1} \binom{p+1}{j} \mathbb{E}X^j\, B_{p+1-j}(1)\right]. \tag{1.25}$$

Proof. Set $A(t) = \int_0^1 \mathrm{up}(x)\mathrm{e}^{tx}\,\mathrm{d}x$. Evenness of up gives $A(t) + A(-t) = \mathsf{M}(t)$, while the partition identity $\mathrm{up}(x) + \mathrm{up}(1-x) = 1$ on $[0,1]$ (a restatement of (1.3) and (1.5)) gives $A(t) + \mathrm{e}^t A(-t) = (\mathrm{e}^t - 1)/t$. Solving the linear system, $A(-t) = 1/t - \mathsf{M}(t)/(\mathrm{e}^t - 1)$. By (1.3), $\int_0^1 F(x)\mathrm{e}^{tx}\,\mathrm{d}x = \mathrm{e}^t A(-t)$, which is (1.24). Expanding $t\mathrm{e}^t/(\mathrm{e}^t - 1) = \sum_n B_n(1)t^n/n!$ and $\mathsf{M}(t) = \sum_j \mathbb{E}X^j\, t^j/j!$ gives (1.25). $\square$

The positive-half Rvachev moments follow from the same computation:

$$a_r := \int_0^1 x^r\, \mathrm{up}(x)\,\mathrm{d}x = \frac{1}{r+1} - I_r = \frac{d_{r+1}}{r+1}, \tag{1.26}$$

where the last form uses (1.21); equivalently the absolute Rvachev moments are

$$\int_{-1}^1 |x|^\alpha\, \mathrm{up}(x)\,\mathrm{d}x = \frac{2d_{\alpha+1}}{\alpha+1} \qquad (\Re\alpha > -1), \tag{1.27}$$

an identity that continues meromorphically in section 8.3.

1.4 Exact dyadic values: one identity, three algorithms

Every sample used in this volume is an exact rational number. The repository's dyadic evaluator can be organized as a single convolution identity with three computationally different specializations; we state it with a self-contained probabilistic proof because both parts of this volume lean on the structure of the proof, not only on the statement.

Define the *Appell moment polynomial* and the *dyadic prefactor*

$$G_m(y) := \mathbb{E}\,(y - X)^m = \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m}{2j} c_j\, y^{m-2j}, \qquad C_m := \frac{2^{-m(m+1)/2}}{m!}. \tag{1.28}$$

G_m is monic of degree m, even or odd according to the parity of m, with $G_m' = mP_{m-1}$ (an Appell sequence for the law of X).

Theorem 1.2 (Dyadic Appell–Thue–Morse convolution). *For every $m \in \mathbb{N}_0$ and $0 \le k \le M = 2^m$,*

$$\boxed{F\Big(\frac{k}{M}\Big) = C_m \sum_{a=0}^{k-1} \varepsilon_a \, G_m\big(2(k-a)-1\big),} \tag{1.29}$$

the empty sum at $k=0$ being zero.

Proof. Split the series (1.1) for X after $n = m+1$ terms and rescale:

$$2^n X = \sum_{\nu=1}^{n} 2^{n-\nu} V_\nu + X',$$

where X' is an independent copy of X. The finite sum is an Irwin–Hall-type variable: a sum of independent uniforms on $[-2^{n-\nu}, 2^{n-\nu}]$, $\nu = 1, \dots, n$. Because the half-widths $1, 2, \dots, 2^{n-1}$ have subset sums exhausting $\{0, 1, \dots, 2^n - 1\}$ *without repetition*, inclusion–exclusion over the 2^n sign choices collapses to a Thue–Morse–signed spline: the density of the finite sum is

$$f_n(y) = \frac{2^{-n(n+1)/2}}{(n-1)!} \sum_{a=0}^{2^n-1} \varepsilon_a \, (y + 2^n - 1 - 2a)_+^{n-1}.$$

The self-similarity $2^{-n}\,\mathrm{up}(y/2^n) = \mathbb{E} f_n(y - X')$ of the scaled density, evaluated at the dyadic point $y = 2(k-M)$ and combined with $\mathrm{up}(k/M - 1) = F(k/M)$ from (1.3), turns each truncated power into the expectation $\mathbb{E}(2(k-a) - 1 - X')^m = G_m(2(k-a)-1)$ precisely for $a \le k-1$ (for larger a the truncated power vanishes because $|X'| \le 1$), and the constants assemble to C_m. □

Example 1.3. For $m = 2$: $P_2(y) = y^2 + \frac{1}{9}$, $C_2 = \frac{1}{16}$, and $F(1/4) = C_2\, P_2(1) = \frac{1}{16}\big(1 + \frac{1}{9}\big) = \frac{5}{72}$; for $m = 3$ the complete row is

$$\big(F(k/8)\big)_{k=0}^{8} = \Big(0, \tfrac{1}{288}, \tfrac{5}{72}, \tfrac{73}{288}, \tfrac{1}{2}, \tfrac{215}{288}, \tfrac{67}{72}, \tfrac{287}{288}, 1\Big).$$

Reflection $F(k/M) + F((M-k)/M) = 1$ is visible and is used as a standing self-check in every experiment of this volume.

Three specializations of (1.29) serve different purposes.

(a) Streaming the complete row. Expanding $G_m(2z+1) = 2^m \sum_j \binom{m}{j} d_j \, z^{m-j}$ (the binomial transform connecting (1.28) to the Fabius-law moments (1.19)) turns (1.29) into

$$F\Big(\frac{a}{2^m}\Big) = \frac{2^{-\binom{m}{2}}}{m!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{j=0}^{m} \binom{m}{j} d_j \, (a-1-h)^{m-j}. \tag{1.30}$$

Algorithm 1.4 (Streaming exact comb). Maintain the signed prefix power sums $S_k(a) = \sum_{h=0}^{a-1} \varepsilon_h (a - 1 - h)^k$ for $0 \le k \le m$ via

$$S_0(a+1) = S_0(a) + \varepsilon_a, \qquad S_k(a+1) = \sum_{q=0}^{k} \binom{k}{q} S_q(a) \quad (k \ge 1),$$

and output $F(a/2^m) = 2^{-\binom{m}{2}} (m!)^{-1} \sum_j \binom{m}{j} d_j S_{m-j}(a)$. The complete level-$m$ row costs $\mathcal{O}(Mm^2)$ exact rational operations. An equivalent sparse form multiplies the coefficient array of $\sum_{d\ge1} G_m(2d-1) z^d$ successively by $(1-z), (1-z^2), \dots, (1-z^{2^{m-1}})$; each factor is one backward difference, for $\mathcal{O}(mM)$ operations total.

(b) Single values by bit recursion. With $V_n := F(2^{-n})$, conditioning on the leading binary digit gives

$$V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}} V_k}{(n-k+1)!}, \qquad F\left(\frac{a}{2^e}\right) = \sum_{j=0}^{r} \frac{2^{\binom{j+1}{2}} V_{r-j}}{j!} \left(\frac{q}{2^e}\right)^j - F\left(\frac{q}{2^e}\right), \tag{1.31}$$

where $2^b \le a < 2^{b+1}$, $r = e - b$, $q = a - 2^b$. This computes one value in $\mathcal{O}(e^2)$ operations and underlies the classical arithmetic of Arias de Reyna and Haugland [6, 7].

(c) Symbolic convolution. Identity (1.29) itself, read as a coefficient convolution, is the input to the generating-function calculus of section 3.

All experiments behind this volume implement at least two of the three routes and compare them exactly; see section 11 and section 18.

Part I
Comb sums and quadrature

2 The dyadic combs

2.1 Left, right, and trapezoidal Fabius combs

For $\alpha \in \mathbb{C}$ define the *punctured left Fabius comb*

$$L_{m,\alpha} := h \sum_{k=1}^{M-1} \left(\frac{k}{M}\right)^{\alpha} F\left(\frac{k}{M}\right), \tag{2.1}$$

with the real logarithm on $(0, 1]$. For $\alpha = p \in \mathbb{N}_0$ this agrees with the ordinary left Riemann sum of $x^p F(x)$ including $k = 0$, since $F(0) = 0$; the puncture is the correct convention for negative and nonintegral powers, where flatness of F at 0 makes even that convention innocuous. At fixed m, $\alpha \mapsto L_{m,\alpha}$ is a finite exponential sum, hence *entire*.

The right and trapezoidal combs differ only through the endpoint $F(1) = 1$:

$$R_{m,\alpha} = L_{m,\alpha} + h, \qquad T_{m,p} = L_{m,p} + \frac{h}{2}. \tag{2.2}$$

The trapezoidal normalization is the natural one for Fourier aliasing (the periodic extension of $x^p F(x)$ has a unit jump at the integers, and its Fourier series converges to the midpoint value there); the right rule R produces the cleanest single-formula closed forms (theorem 5.5); the left rule L matches the generated tables of the absorbed sources. All statements below convert freely via (2.2).

2.2 Shifted and midpoint combs

For a phase $\theta \in \mathbb{R}$ define

$$L^{(\theta)}_{m,\alpha} := h \sum_{\substack{k \in \mathbb{Z} \\ 0 < k+\theta < M}} \left(\frac{k+\theta}{M}\right)^{\alpha} F\left(\frac{k+\theta}{M}\right). \tag{2.3}$$

A dyadic phase is a residue-class extraction from a finer level: the midpoint comb, for instance, satisfies the exact two-scale identity

$$L^{(1/2)}_{m,p} = 2L_{m+1,p} - L_{m,p}, \tag{2.4}$$

because the level-$(m+1)$ nodes split into the level-m nodes and the midpoints. More generally:

Proposition 2.1 (Dyadic phase as a roots-of-unity filter). *Let $\theta = a/2^s$ with $0 \le a < 2^s$, put $D = m+s$ and $\omega = \exp(2\pi i/2^s)$, and let A_D be the row polynomial of theorem 3.1 at level D. Then, with the spectral Euler power* (3.11),

$$L_{m,\alpha}^{(a/2^s)} = \frac{2^{-D\alpha}}{M\,2^s} \sum_{r=0}^{2^s-1} \omega^{-ar}\,(\mathcal{D}^\alpha A_D)(\omega^r). \tag{2.5}$$

Proof. The character projector $2^{-s}\sum_r \omega^{r(k-a)}$ selects the indices $k \equiv a \pmod{2^s}$ from the level-D row, and $\mathcal{D}^\alpha$ implements the power weight coefficientwise. ☐

Every standard one-dimensional quadrature convention on the dyadic comb is therefore an exact linear functional of finitely many objects $(\mathcal{D}^\alpha A_D)(\zeta)$ at 2-power roots of unity ζ. This is the first indication that the *row generating function* is the right primitive object; we develop it next.

2.3 Rvachev combs

The corresponding two-sided objects, studied in sections 7 and 8, are the shifted monomial combs

$$Q_{m,p}(\theta) := h \sum_{k\in\mathbb{Z}} \big(h(k+\theta)\big)^p \,\mathrm{up}\big(h(k+\theta)\big), \tag{2.6}$$

finite sums because $\operatorname{supp}\mathrm{up} = [-1,1]$, and the punctured absolute combs

$$U_m(\alpha) := h \sum_{k\in\mathbb{Z}\setminus\{0\}} |kh|^\alpha \,\mathrm{up}(kh). \tag{2.7}$$

On $[0,1]$ the fold (1.3) links them to the Fabius combs; with the pure power comb $\Pi_{m,\alpha} := h\sum_{k=1}^{M-1}(k/M)^\alpha = M^{-\alpha-1}[\zeta(-\alpha) - \zeta(-\alpha, M)]$,

$$U_m(\alpha) = 2\big(\Pi_{m,\alpha} - L_{m,\alpha}\big) \qquad (\alpha \in \mathbb{C}), \tag{2.8}$$

so every Fabius-comb theorem has an immediate Rvachev shadow and conversely. This elementary bridge carries surprising weight: in section 7 it converts the exact Euler–Maclaurin collapse into exact quadrature statements, and in section 8 it transports the Hurwitz-zeta calculus to the two-sided combs.

3 The rational row generating function

3.1 Pole collapse

Encode a complete level-m sample row in the ordinary generating function

$$A_m(z) := \sum_{k=0}^{M-1} F(k/M)\, z^k. \tag{3.1}$$

Let $\mathcal{D} = z\,\mathrm{d}/\mathrm{d}z$ be the Euler operator and recall the finite Thue–Morse product $\Theta_m(z) = \prod_{j=0}^{m-1}(1 - z^{2^j}) = \sum_{a=0}^{M-1} \varepsilon_a z^a$.

Theorem 3.1 (Complete sample-row generating function). *Define*

$$\mathscr{F}_m(z) := C_m\,\Theta_m(z)\,G_m(2\mathcal{D} - 1)\,\frac{z}{1-z}. \tag{3.2}$$

Then $\mathscr{F}_m$ is a rational function with a single simple pole at $z = 1$, and

$$\mathscr{F}_m(z) = \sum_{k\ge 0} \widetilde{F}_m(k) z^k, \qquad \widetilde{F}_m(k) = \begin{cases} F(k/M), & 0 \le k \le M, \\ 1, & k \ge M. \end{cases} \tag{3.3}$$

In particular

$$\boxed{A_m(z) = \mathscr{F}_m(z) - \frac{z^M}{1-z}} \tag{3.4}$$

is a polynomial of degree at most $M-1$.

Proof. The kernel $\sum_{r\geq 1} G_m(2r-1)z^r = G_m(2\mathcal{D}-1)\,z/(1-z)$ has coefficient $G_m(2r-1)$ at z^r, so the coefficient of z^k in (3.2) is exactly the convolution (1.29); this proves the coefficient identity for $0 \leq k \leq M$. For the pole structure, each application of $(2\mathcal{D}-1)$ to a rational function with denominator $(1-z)^{r+1}$ raises the denominator order by one, so $G_m(2\mathcal{D}-1)z/(1-z)$ has denominator $(1-z)^{m+1}$. On the other hand,

$$\Theta_m(z) = (1-z)^m \prod_{j=0}^{m-1}\left(1+z+\cdots+z^{2^j-1}\right), \tag{3.5}$$

so the finite Thue–Morse mask cancels exactly m of the $m+1$ poles, leaving one simple pole; the numerator degree count $(M-1)+(m+1)-m = M$ shows that the coefficients of $\mathscr{F}_m$ are constant from index M on. The coefficient at index M is $F(1)=1$, so the constant tail is 1, and subtracting the tail $z^M/(1-z)$ leaves the polynomial A_m. □

Remark 3.2 (What the pole collapse says). Prouhet cancellation is usually quoted as $\sum_a \varepsilon_a\, a^r = 0$ for $r < m$. Theorem 3.1 is its generating-function upgrade: the Thue–Morse mask annihilates the entire order-m polar part generated by the degree-m moment kernel, and the surviving simple pole encodes precisely the saturation $F \equiv 1$ past the right endpoint. The low levels are

$$\mathscr{F}_1(z) = \frac{z(z+1)}{2(1-z)}, \tag{3.6}$$

$$\mathscr{F}_2(z) = \frac{z(z+1)(z+5)(5z+1)}{72(1-z)}, \tag{3.7}$$

$$\mathscr{F}_3(z) = \frac{z(z+1)^3(z^2+1)(z^2+16z+1)}{288(1-z)}. \tag{3.8}$$

After removing the initial factor z, the numerators are reciprocal polynomials — an exact reflection of $F(k/M) + F((M-k)/M) = 1$, made precise by the functional equation

$$A_m(z) + z^M A_m(1/z) = \frac{z\,(1-z^{M-1})}{1-z}, \tag{3.9}$$

proved in Appendix C. A factorization theory for these row numerators (cyclotomic multiplicities, the limiting law of the noncyclotomic zeros) is posed as theorem 12.1.

3.2 Comb sums as Euler derivatives of the row

Corollary 3.3 (All powers from one polynomial)**.** *For every* $p \in \mathbb{N}_0$*,*

$$L_{m,p} = M^{-p-1}\,(\mathcal{D}^p A_m)(1), \tag{3.10}$$

and with the spectral Euler power

$$\mathcal{D}^\alpha\Big(\sum_{k\geq 0} a_k z^k\Big) := \sum_{k\geq 1} k^\alpha a_k z^k \qquad (\alpha \in \mathbb{C}) \tag{3.11}$$

the same formula $L_{m,\alpha} = M^{-\alpha-1}(\mathcal{D}^\alpha A_m)(1)$ *holds for every complex exponent.*

Together with theorem 2.1 this reduces *every* comb convention of section 2 to coefficient extractions from A_m — a picture completed by the closed forms of the next section, which eliminate the 2^m-term row itself.

4 Closed forms for nonnegative integer powers

4.1 The Bernoulli–Thue–Morse formula

Inserting the convolution (1.29) into (2.1) and summing the resulting power sums by Faulhaber's formula $\sum_{r=1}^{K} r^q = \big[B_{q+1}(K+1) - B_{q+1}(1)\big]/(q+1)$ gives an exact finite formula at every level and degree.

Theorem 4.1 (Exact closed form for every m, p)**.** *For $p \in \mathbb{N}_0$,*

$$L_{m,p} = C_m M^{-p-1} \sum_{a=0}^{M-1} \varepsilon_a \, H_{m,p}(a), \tag{4.1}$$

where

$$H_{m,p}(a) = \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m}{2j} c_j \sum_{r=0}^{p} \binom{p}{r} a^{p-r} \sum_{s=0}^{m-2j} \binom{m-2j}{s} 2^s (-1)^{m-2j-s} \frac{B_{r+s+1}(M-a) - B_{r+s+1}(1)}{r+s+1}. \tag{4.2}$$

Proof. By (1.29),

$$M^{p+1} C_m^{-1} L_{m,p} = \sum_{a=0}^{M-2} \varepsilon_a \sum_{r=1}^{M-1-a} (a+r)^p \, G_m(2r-1).$$

Expand $(a+r)^p$ binomially, expand $(2r-1)^d$ for each monomial of G_m, and apply Faulhaber with $K = M - 1 - a$. □

Formula (4.1) is manifestly rational and is one of the three independent exact routes used in the verification suite (section 11). It still contains a complete 2^m-term signed block; the next theorem shows that only a bounded number of its coefficients can ever matter.

4.2 Bell–Prouhet collapse of the signed block

Define the signed Thue–Morse power moments

$$T_{m,n} := \sum_{a=0}^{M-1} \varepsilon_a \, a^n, \tag{4.3}$$

whose exponential generating function is the finite product $\sum_n T_{m,n} t^n/n! = \Theta_m(\mathrm{e}^t) = \prod_{j=0}^{m-1} (1 - \mathrm{e}^{2^j t})$.

Theorem 4.2 (Bell formula for complete Thue–Morse power blocks)**.** *Let*

$$\lambda_{m,1} = \frac{M-1}{2}, \qquad \lambda_{m,2r} = \frac{B_{2r}}{2r} \frac{2^{2rm}-1}{2^{2r}-1} \quad (r \ge 1), \qquad \lambda_{m,2r+1} = 0 \quad (r \ge 1). \tag{4.4}$$

Then

$$\Theta_m(\mathrm{e}^t) = (-1)^m 2^{m(m-1)/2} \, t^m \exp\Big(\sum_{r \ge 1} \lambda_{m,r} \frac{t^r}{r!}\Big), \tag{4.5}$$

whence $T_{m,n} = 0$ for $n < m$ (Prouhet cancellation) and, for $s \ge 0$,

$$\boxed{T_{m,m+s} = (-1)^m 2^{m(m-1)/2} \frac{(m+s)!}{s!} Y_s(\lambda_{m,1}, \ldots, \lambda_{m,s}).} \tag{4.6}$$

Proof. Factor $1-\mathrm{e}^x = -x\,\mathrm{e}^{x/2}\,\dfrac{\sinh(x/2)}{x/2}$ and use the classical expansion $\log\big[(\mathrm{e}^x-1)/x\big] = x/2 + \sum_{r\geq 1}\frac{B_{2r}}{2r}\frac{x^{2r}}{(2r)!}$, summed over $x = 2^j t$, $0\leq j<m$: the prefactors contribute $(-1)^m 2^{0+1+\cdots+(m-1)}t^m$, the linear terms sum to $(M-1)t/2$, and each even term is the finite geometric sum in (4.4). Comparing $\exp(\sum_r x_r t^r/r!) = \sum_s Y_s(x_1,\dots,x_s)t^s/s!$ with the definition of $T_{m,n}$ gives (4.6). □

Corollary 4.3 (Only $p+2$ signed moments survive)**.** *The polynomial $H_{m,p}$ of (4.2) has degree at most $m+p+1$, so*

$$\boxed{L_{m,p} = C_m M^{-p-1}\sum_{s=0}^{p+1}\big[a^{m+s}\big]H_{m,p}(a)\;T_{m,m+s},} \tag{4.7}$$

with $T_{m,m+s}$ given by (4.6). The full 2^m-term Thue–Morse block collapses to at most $p+2$ Bell-polynomial terms, uniformly in m.

Proof. In (4.2) the Bernoulli polynomial has degree $r+s+1$ in a after the substitution $a\mapsto M-a$; multiplied by a^{p-r} the total degree is at most $p+s+1\leq m+p+1$. Monomials of degree below m are annihilated by (4.6). □

A computational dichotomy

For fixed m and many degrees p, stream the row once (theorem 1.4, $\mathcal{O}(mM)$ operations) and read off every $L_{m,p}$ from theorem 3.3. For fixed p and large or symbolic m, use the Bell collapse: its signed part has constant length $p+2$, and all dependence on the exponentially long block is compressed into the explicit jets $\lambda_{m,r}$ — a closed form in which m appears only through 2^m. The two routes, together with the direct sample sum, are compared exactly in section 11.

Remark 4.4 (Eulerian form of the kernel)*.* For symbolic work in z rather than t, the kernel of theorem 3.1 expands through negative-index polylogarithms,

$$\sum_{d\geq 1} G_m(2d-1)z^d = \sum_{j=0}^{\lfloor m/2\rfloor}\binom{m}{2j}c_j\sum_{i=0}^{m-2j}\binom{m-2j}{i}2^i(-1)^{m-2j-i}\,\mathrm{Li}_{-i}(z), \tag{4.8}$$

each $\mathrm{Li}_{-i}(z) = z\,\mathcal{A}_i(z)/(1-z)^{i+1}$ rational with Eulerian numerator $\mathcal{A}_i$. This makes (3.2) effective in any computer algebra system without operator calculus.

5 The exact Euler–Maclaurin collapse

For a smooth integrand, the Euler–Maclaurin formula is an asymptotic expansion with a remainder that is small but almost never zero. For $x^pF(x)$ on the dyadic comb the situation is remarkably better: the expansion *terminates and becomes an exact identity* as soon as the level clears an explicit threshold linear in p. This section proves the collapse, identifies the sharp-looking threshold, and prepares the exact description of the first failure.

5.1 Trapezoidal aliasing through the characteristic function

Put $g_p(x) = x^pF(x)$ on $[0,1]$ and $\widehat{g}_p(n) = \int_0^1 g_p(x)\mathrm{e}^{-2\pi i n x}\,\mathrm{d}x$. The periodic extension of g_p has a unit jump at the integers, and its Fourier series converges there to the midpoint value, which is exactly the trapezoidal endpoint convention:

$$T_{m,p} = \sum_{\ell\in\mathbb{Z}}\widehat{g}_p(M\ell), \tag{5.1}$$

the zero-frequency term being I_p. Every nonzero coefficient is a finite expression in the Fabius characteristic function (1.9).

Lemma 5.1 (Fourier coefficients of $x^pF(x)$). *For $n \in \mathbb{Z} \setminus \{0\}$,*

$$\widehat{g}_p(n) = \sum_{j=0}^{p} \frac{p^{\underline{j}}}{(2\pi \mathrm{i} n)^{j+1}} \left(\frac{\chi^{(p-j)}(n)}{(-2\pi\mathrm{i})^{p-j}} - 1 \right). \tag{5.2}$$

In particular, for $p = 0$,

$$\widehat{g}_0(\xi) = -\frac{\mathrm{e}^{-2\pi\mathrm{i}\xi}}{2\pi\mathrm{i}\xi} + \frac{\chi(\xi)}{2\pi\mathrm{i}\xi} \qquad (\xi \neq 0), \tag{5.3}$$

valid for all real ξ, not only integers.

Proof. Since $F(x) = \mathbb{P}(Y \leq x)$, Fubini gives the layer representation $\widehat{g}_p(n) = \mathbb{E}\int_Y^1 x^p \mathrm{e}^{-2\pi\mathrm{i}nx}\,\mathrm{d}x$. Integrating by parts $p+1$ times, using $\mathrm{e}^{-2\pi i n} = 1$ at the fixed endpoint,

$$\int_Y^1 x^p \mathrm{e}^{-2\pi\mathrm{i}nx}\,\mathrm{d}x = \sum_{j=0}^{p} \frac{p^{\underline{j}}}{(2\pi\mathrm{i}n)^{j+1}} \Big(Y^{\,p-j}\mathrm{e}^{-2\pi\mathrm{i}nY} - 1 \Big),$$

and $\mathbb{E}\big[Y^r \mathrm{e}^{-2\pi\mathrm{i}nY}\big] = \chi^{(r)}(n)/(-2\pi\mathrm{i})^r$. For (5.3), one integration by parts of $\int_0^1 F\,\mathrm{e}^{-2\pi\mathrm{i}\xi x}\,\mathrm{d}x$ with $F' = \rho$, $\widehat{\rho} = \chi$ suffices. $\square$

Splitting (5.2) into its "-1" part (a pure boundary term) and its χ part (a spectral term) splits (5.1) accordingly. The boundary part sums by Euler's evaluation of $\zeta(2r)$ into the classical Bernoulli corrections; the spectral part is an exact, absolutely convergent alias remainder.

Theorem 5.2 (Euler–Maclaurin with exact alias remainder). *For every $m \geq 0$ and $p \in \mathbb{N}_0$,*

$$L_{m,p} = I_p - \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!}\, p^{\underline{2r-1}}\, h^{2r} + \mathcal{R}_{m,p}, \tag{5.4}$$

where

$$\mathcal{R}_{m,p} = \sum_{\ell \in \mathbb{Z}\setminus\{0\}} \sum_{j=0}^{p} \frac{p^{\underline{j}}}{(2\pi\mathrm{i}M\ell)^{j+1}} \frac{\chi^{(p-j)}(M\ell)}{(-2\pi\mathrm{i})^{p-j}}. \tag{5.5}$$

Proof. Insert (5.2) into (5.1). In the boundary part only odd $j = 2r-1$ survives the symmetric ℓ-sum, and

$$-\frac{p^{\underline{2r-1}}}{(2\pi\mathrm{i}M)^{2r}} \sum_{\ell\neq 0} \ell^{-2r} = \frac{B_{2r}}{(2r)!}\, p^{\underline{2r-1}} M^{-2r}$$

by $\zeta(2r) = (-1)^{r+1}B_{2r}(2\pi)^{2r}/(2(2r)!)$. Finally $T_{m,p} = L_{m,p} + h/2$. $\square$

5.2 The collapse and its threshold

Define the parity threshold

$$\tau(0) := 0, \qquad \tau(p) := 2\lceil p/2 \rceil = \begin{cases} p, & p \text{ even}, \\ p+1, & p \text{ odd}, \end{cases} \qquad p \geq 1; \tag{5.6}$$

equivalently, $m \geq \tau(p)$ if and only if $p \leq 2\lfloor m/2 \rfloor$.

Theorem 5.3 (Exact Euler–Maclaurin collapse). *If $m \geq \tau(p)$, then $\mathcal{R}_{m,p} = 0$; that is,*

$$\boxed{L_{m,p} = I_p - \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!}\, p^{\underline{2r-1}}\, h^{2r}.} \tag{5.7}$$

Equivalently, the level-m comb agrees exactly with its finite Euler–Maclaurin polynomial through degree m when m is even, and through degree $m-1$ when m is odd.

Proof. For $m = 0$, $p = 0$ the identity reads $L_{0,0} = 0 = I_0 - \frac{1}{2}$, which is the symmetry value $I_0 = \int_0^1 F = \frac{1}{2}$. Let $m \geq 1$. By (1.12), χ vanishes at $M\ell$ to order $m + \nu_2(\ell) \geq m$. If $p < m$, every derivative order $p - j \leq p$ appearing in (5.5) is below the zero order, so the remainder vanishes term by term.

It remains to treat $p = m$ for even m. Aliases with even ℓ still vanish ($\nu_2(\ell) \geq 1$), and at odd ℓ only the $j = 0$ term, carrying the derivative order exactly m, can survive. From $\chi(\xi) = \mathrm{e}^{-\pi \mathrm{i}\xi}\Phi(\xi/2)$ and the vanishing of all lower derivatives of $\Phi(\cdot/2)$ at $M\ell$, the Leibniz rule leaves a single term,

$$\chi^{(m)}(M\ell) = \mathrm{e}^{-\pi \mathrm{i} M\ell}\, 2^{-m}\, \Phi^{(m)}\left(\frac{M\ell}{2}\right), \qquad \mathrm{e}^{-\pi \mathrm{i} M\ell} = 1 \;\; (m \geq 1).$$

Since Φ is even, $\Phi^{(m)}$ is even for even m, so $\chi^{(m)}(M\ell)$ is then even in ℓ, whereas the accompanying factor $(2\pi \mathrm{i} M\ell)^{-1}$ is odd; the aliases ℓ and $-\ell$ cancel in pairs. For odd m no cancellation is forced, and theorem 6.1 shows the surviving sum is generically nonzero. This is precisely the range $m \geq \tau(p)$. □

Remark 5.4 (Why this is not ordinary polynomial exactness). The summand contains the non-polynomial factor F. The theorem does not assert that a fixed rule integrates polynomials against a fixed weight; it asserts that the *nonzero Fourier aliases of the particular integrand* are annihilated by the growing zeros of χ at the dual lattice. It is a spectral cancellation theorem for a family of ordinary Riemann sums — a Strang–Fix phenomenon [12, 13] in which the reproduced polynomial degree grows with the refinement level because each refinement contributes a fresh vanishing sinc factor.

5.3 One-formula closed form and low-degree examples

Because the finite Euler–Maclaurin polynomial is itself a Bernoulli polynomial in disguise, the collapse compresses to a single formula.

Corollary 5.5 (Exact right-comb formula). *For $p \in \mathbb{N}_0$ and $m \geq \tau(p)$,*

$$\boxed{R_{m,p} = \frac{h^{p+1}B_{p+1}(M+1) - d_{p+1}}{p+1}.} \tag{5.8}$$

Proof. Expanding $B_{p+1}(M+1)$ around $M = 1/h$,

$$\frac{h^{p+1}B_{p+1}(M+1) - 1}{p+1} = \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!} p^{\underline{2r-1}} h^{2r},$$

which combined with $I_p = (1 - d_{p+1})/(p+1)$ and $R_{m,p} = L_{m,p} + h$ is (5.7). □

The Bernoulli polynomial encodes the pure Faulhaber sum, and d_{p+1} carries all the Fabius arithmetic: the exact right comb is *the power sum minus one Fabius-law moment.* The first cases of (5.7), generated and exactly verified by the companion script of the absorbed sums package (`assets/Fabius_Dyadic_Comb_Sums_Report_Package/`), are:

p	I_p	threshold $\tau(p)$	exact $L_{m,p}$ for $m \geq \tau(p)$
0	$\frac{1}{2}$	0	$\frac{1}{2} - \frac{h}{2}$
1	$\frac{13}{36}$	2	$\frac{13}{36} - \frac{h}{2} + \frac{1}{12}h^2$
2	$\frac{5}{18}$	2	$\frac{5}{18} - \frac{h}{2} + \frac{1}{6}h^2$
3	$\frac{1207}{5400}$	4	$\frac{1207}{5400} - \frac{h}{2} + \frac{1}{4}h^2 - \frac{1}{120}h^4$
4	$\frac{251}{1350}$	4	$\frac{251}{1350} - \frac{h}{2} + \frac{1}{3}h^2 - \frac{1}{30}h^4$
5	$\frac{22661}{142884}$	6	$\frac{22661}{142884} - \frac{h}{2} + \frac{5}{12}h^2 - \frac{1}{12}h^4 + \frac{1}{252}h^6$
6	$\frac{16427}{119070}$	6	$\frac{16427}{119070} - \frac{h}{2} + \frac{1}{2}h^2 - \frac{1}{6}h^4 + \frac{1}{42}h^6$
7	$\frac{63446897}{520506000}$	8	$\frac{63446897}{520506000} - \frac{h}{2} + \frac{7}{12}h^2 - \frac{7}{24}h^4 + \frac{1}{12}h^6 - \frac{1}{240}h^8$
8	$\frac{7096441}{65063250}$	8	$\frac{7096441}{65063250} - \frac{h}{2} + \frac{2}{3}h^2 - \frac{7}{15}h^4 + \frac{2}{9}h^6 - \frac{1}{30}h^8$

Table 1: Exact terminating Euler–Maclaurin identities. Each row is an identity of rational functions of $h = 2^{-m}$, valid for every level $m \geq \tau(p)$; below the threshold the defect of section 6 appears.

The complete exactness pattern through level 8 was verified in exact rational arithmetic (three independent routes; see section 11) and matches $\{(m, p) : m \geq \tau(p)\}$ perfectly — including the *first* failures, which is the subject of the next section.

6 The first defect and spectral Dirichlet values

Below the threshold $\tau(p)$ the alias remainder (5.5) is nonzero — but it is not an opaque error term. At the first failing degree it is a single, explicitly evaluable Dirichlet series over the half-integer values of the sinc product, and the exactness of everything else turns that series into a machine for producing closed forms.

6.1 The odd-level defect series

Theorem 6.1 (First odd-level defect)**.** *Let $m \geq 1$ be odd. Then*

$$\boxed{\mathcal{R}_{m,m} = \frac{2\,(-1)^{(m+1)/2}\,m!}{(2\pi)^{m+1}\,2^{m(m+1)}}\,D(m+1), \qquad D(s) := \sum_{\substack{q\geq 1\\ q \text{ odd}}} \frac{\Phi(q/2)}{q^s},} \tag{6.1}$$

where D is entire in s by the rapid decay of Φ. Moreover the summands obey the Thue–Morse sign law

$$\operatorname{sgn}\Phi\left(\frac{2t+1}{2}\right) = \varepsilon_t \qquad (t \geq 0). \tag{6.2}$$

Proof. For $p = m$ odd, the proof of theorem 5.3 leaves in (5.5) only the $j = 0$ term at odd ℓ, with

$$\chi^{(m)}(M\ell) = 2^{-m}\,\Phi^{(m)}\big(2^{m-1}\ell\big) = -\,2^{-m}\,\frac{m!}{(2^{m-1}\ell)^m}\,\Phi\left(\frac{\ell}{2}\right)$$

by the leading-jet formula (1.11) at the order-m zero $2^{m-1}\ell$. Substituting into (5.5) and pairing ℓ with $-\ell$ (both terms are equal for odd m) gives (6.1) after collecting powers of 2. The sign law (6.2) follows by counting sign changes of the real even product Φ along $[0, q/2]$: each factor $\operatorname{sinc}_\pi(\xi/2^j)$ changes sign at the odd multiples of 2^j, and the total number of integer zeros (with multiplicity) in $(0, q/2)$ crossed by the product has the parity of the binary digit sum of $t = (q-1)/2$; alternatively, peel one factor as in (6.3) below and induct on t. □

Two structural identities frame D. Peeling the $j = 0$ factor of the product gives the *half-scale relation*

$$\Phi\left(\frac{q}{2}\right) = \operatorname{sinc}_\pi\left(\frac{q}{2}\right)\Phi\left(\frac{q}{4}\right) = \frac{2\,(-1)^{(q-1)/2}}{\pi q}\,\Phi\left(\frac{q}{4}\right) \qquad (q \text{ odd}), \tag{6.3}$$

which connects $D(s)$ to a twisted quarter-integer series and is the natural starting point for the functional-equation program of theorem 6.7. And since $\Phi(q/2) \to 0$ rapidly as $q \to \infty$ while the $q = 1$ term is fixed,

$$D(s) = \Phi(\tfrac{1}{2}) + \Phi(\tfrac{3}{2})\,3^{-s} + \mathcal{O}(5^{-s}) \qquad (s \to \infty), \tag{6.4}$$

with $\Phi(\frac{1}{2}) = 0.5537712\ldots > 0$ and $\Phi(\frac{3}{2}) = -0.0462022\ldots < 0$: the values $D(s)$ increase to the limit $\Phi(1/2)$ from below.

6.2 Exact rational multiples of powers of π

The defect identity can be read in both directions. Read forward, it evaluates the first comb defect; read backward, it evaluates the Dirichlet series, because everything else in (5.4) is an explicitly computable rational number.

Theorem 6.2 (Spectral rationality)**.** *For every $r \geq 1$,*

$$\boxed{\frac{D(2r)}{\pi^{2r}} \in \mathbb{Q},} \qquad \textit{effectively: } D(2r) = (-1)^r \frac{2^{(2r)^2}}{2\,(2r-1)!}\,\mathcal{R}_{2r-1,\,2r-1}\,\pi^{2r}, \tag{6.5}$$

where $\mathcal{R}_{2r-1,2r-1} = L_{2r-1,2r-1} - \big[$the finite Euler–Maclaurin polynomial of (5.7)$\big]$ is an exact rational number computable from the level-$(2r-1)$ comb.

Proof. Solve (6.1) for $D(m+1)$ with $m = 2r-1$; the prefactor is $(-1)^{(m+1)/2}(2\pi)^{m+1}2^{m(m+1)}/(2\,m!) = (-1)^r 2^{(m+1)^2}\pi^{m+1}/(2\,m!)$. The defect on the right of (6.1) is rational by theorem 4.1 and (1.21). ☐

Carrying this out with exact arithmetic gives the first four values:

$$\boxed{D(2) = \frac{\pi^2}{18}, \qquad D(4) = \frac{23\,\pi^4}{4050}, \qquad D(6) = \frac{1543\,\pi^6}{2679075}, \qquad D(8) = \frac{1196113\,\pi^8}{20494923750}.} \tag{6.6}$$

Numerically $0.548311\ldots$, $0.553187\ldots$, $0.553707\ldots$, $0.553764\ldots$, converging to $\Phi(1/2) = 0.553771\ldots$ as (6.4) predicts. The same route through levels 9 and 11 gives two further exact values,

$$D(10) = \frac{2045676559\,\pi^{10}}{345944065438125}, \qquad D(12) = \frac{1158575464019117\,\pi^{12}}{193371489397785135937 5}, \tag{6.7}$$

numerically $0.5537704925659628\ldots$ and $0.5537711889156489\ldots$. The values $D(2)$ and $D(4)$ were derived in one absorbed source and re-derived here; $D(6)$ through $D(12)$ were computed for this consolidation by an independent implementation of (6.5) (bit-recursion evaluator (1.31), exact `Fraction` arithmetic) that first reproduced all eight first-defect values of table 2 and the complete exactness matrix through level 8, and each closed form was checked against a direct high-precision evaluation of the series to 15 digits.

m	first failing degree p	exact defect $\mathcal{R}_{m,p}$
1	1	$-1/144$
2	3	$-97/1382400$
3	3	$23/22118400$
4	5	$9671/11985978654720$
5	5	$-1543/767102633902080$
6	7	$-9903527/146509414227756711936000$
7	7	$1196113/37506410042305718255616000$
8	9	$21035141003/590082134533477495427314445451264000$

Table 2: Exact first defects through level eight (from the absorbed sums package, independently reproduced for this volume). At odd m the first failing degree is $p = m$ and the defect is (6.1); at even m it is $p = m + 1$.

Remark 6.3 (Even levels and the 2-adic layer structure). At even m the degree-$(m+1)$ defect mixes two contributions: the $j \in \{0, 1\}$ jets of the odd aliases (order-m zeros, now differentiated $m+1$ times) and the leading jet of the $\nu_2(\ell) = 1$ alias layer (order-$(m+1)$ zeros). Each piece is a Dirichlet series of the same half-integer type with explicitly computable local coefficients — the general jet calculus is theorem 7.5 — so the even-level defects in table 2 admit, in principle, the same closed-form treatment. Making that two-layer decomposition arithmetically explicit is part of theorem 6.6.

6.3 Sharpness and positivity

The exactness matrix verified through level 8 shows the first defect occurring exactly at $p = \tau^{-1}$-boundary, and every entry of table 2 is nonzero. The following two conjectures, stated in the absorbed sources in several equivalent forms, remain the central open problems of this part.

Conjecture 6.4 (Sharp threshold). *For every $m \geq 1$ the exactness range of theorem 5.3 is maximal: $\mathcal{R}_{m,p} = 0$ for $p \leq 2\lfloor m/2 \rfloor$ and $\mathcal{R}_{m,p} \neq 0$ at the first degree beyond (i.e. $p = m$ for odd m, $p = m+1$ for even m). Equivalently, no accidental cancellation occurs among the surviving alias jets. Verified exactly for $m \leq 8$.*

Conjecture 6.5 (Spectral positivity). *$D(2r) > 0$ for every $r \geq 1$; equivalently, by (6.5), the odd-level defects satisfy $\operatorname{sgn} \mathcal{R}_{m,m} = (-1)^{(m+1)/2}$. Verified for $r \leq 6$ by the exact values (6.6) and (6.7). The summands alternate in sign by (6.2), so this is not termwise trivial; the first harmonic dominates for large r by (6.4), and the open content is a uniform estimate at small r.*

Conjecture 6.6 (Arithmetic closure of higher alias jets). *After grouping by 2-adic valuation and Fourier parity, every parity-compatible jet series $\sum_{q\,\mathrm{odd}} \Phi^{(j)}(q/2)\, q^{-s}$-combination occurring in an integer comb defect $\mathcal{R}_{m,p}$ lies in $\mathbb{Q}\,\pi^{\,s-j}$. (Rationality of each* total *defect is a theorem, by theorem 4.1; the conjecture asserts that the valuation/jet decomposition is arithmetically diagonal, with no cross-series cancellation needed.)*

Problem 6.7 (Functional equations for D). *Use the half-scale relation (6.3) to split odd residues modulo powers of two and search for a Mahler-type functional equation for the entire function $D(s)$. The exact values (6.6) provide unusually rigid interpolation data; a closed recurrence would in particular settle theorem 6.5.*

7 Shifted Rvachev quadrature and the alias calculus

The two-sided comb sums of up upgrade the Fabius collapse to a quadrature statement that is uniform in an *arbitrary real shift*, with a complete finite calculus for everything the exactness misses.

7.1 Exact polynomial quadrature for every shift

Recall the shifted monomial comb $Q_{m,p}(\theta)$ of (2.6). Shifted Poisson summation for $f_p(x) = x^p \operatorname{up}(x)$, together with $\widehat{f}_p(\xi) = (-2\pi\mathrm{i})^{-p}\Phi^{(p)}(\xi)$, gives the exact absolutely convergent representation

$$Q_{m,p}(\theta) = \int_{-1}^{1} x^p \operatorname{up}(x)\,\mathrm{d}x + \Big(-\frac{1}{2\pi\mathrm{i}}\Big)^p \sum_{\ell\in\mathbb{Z}\setminus\{0\}} \mathrm{e}^{2\pi\mathrm{i}\ell\theta}\,\Phi^{(p)}(M\ell). \tag{7.1}$$

Theorem 7.1 (Shifted dyadic exactness). *For every level $m \in \mathbb{N}_0$, every $\theta \in \mathbb{R}$, and every polynomial P with $\deg P \leq m$,*

$$\boxed{h\sum_{k\in\mathbb{Z}} P\big(h(k+\theta)\big)\,\operatorname{up}\big(h(k+\theta)\big) = \int_{-1}^{1} P(x)\,\operatorname{up}(x)\,\mathrm{d}x.} \tag{7.2}$$

Formal crosswalk. The monomial core of this theorem is kernel-verified: for every real shift θ and every $p \leq m$, `Fabius.tsum_shifted_monomial_eq_integral_real` in `MonomialCombExactness.lean` proves $\sum_{k\in\mathbb{Z}}(\theta+k)^p \operatorname{up}\big(2^{-m}(\theta+k)\big) = \int_{\mathbb{R}} x^p \operatorname{up}(2^{-m}x)\,\mathrm{d}x$ (the display above is this identity multiplied by h^{p+1} and substituted, and extends to polynomials by linearity). The proof is the aliasing argument of this section made formal: `MonomialCombFourier.lean` packages $x^p \operatorname{up}(2^{-m}x)$ as a Schwartz map with $(-2\pi\mathrm{i})^p\widehat{f}_p = (\widehat{f}_0)^{(p)}$, `MonomialCombDerivatives.lean` evaluates $(\widehat{f}_0)^{(p)}$ through the entire transform via the real–complex derivative bridge, `MonomialCombAlias.lean` kills every nonzero alias by the integer-zero order theorem ($\operatorname{ord}_{2^m\ell}\Phi = v_2(2^m\ell)+1 > m$), and Poisson summation for Schwartz maps collapses the comb to the zero-frequency integral. The odd centered moments (theorem 7.2) vanish at every real scale with no level restriction by evenness alone (`Fabius.tsum_odd_pow_mul_rvachevUp` in `DyadicCombTrapezoid.lean`), and the trapezoidal $p=0$ exactness holds on every uniform grid (`Fabius.fabius_trapezoid_exact`).

Proof. By (1.10), $\operatorname{ord}_{\xi=M\ell}\Phi = m+\nu_2(\ell)+1 \geq m+1$ for $\ell \neq 0$, so every alias term in (7.1) vanishes for $p \leq m$; linearity extends monomials to polynomials. □

Corollary 7.2. *For every m and θ, the samples form an exact partition of unity, $h\sum_k \operatorname{up}(h(k+\theta)) = 1$. For the centered comb ($\theta = 0$),*

$$h\sum_k (kh)^{2r}\operatorname{up}(kh) = c_r \quad (2r \leq m), \qquad h\sum_k (kh)^{2r+1}\operatorname{up}(kh) = 0 \quad (\textit{all } m), \tag{7.3}$$

and for $0 < 2r \leq m$ the half-grid identity $h\sum_{k=0}^{M}(kh)^{2r}\operatorname{up}(kh) = c_r/2$ follows by evenness. Tensor products give exact cubature on dyadic lattices in any dimension for all coordinatewise degrees at most m.

Remark 7.3 (Base-b extension). Only the alignment of the sinc zeros with the dual lattice is used. For an integer base $b \geq 2$, the compactly supported density up_b with $\widehat{\operatorname{up}_b}(\xi) = \prod_{j\geq 0}\operatorname{sinc}_\pi(\xi/b^j)$ (an infinite convolution of uniform laws at geometrically decreasing scales) satisfies the same theorem on the comb $h\mathbb{Z}+\theta$, $h = b^{-m}$: at $b^m\ell$ the factors $j = 0,\dots,m$ all vanish. The finer 2-adic valuation structure below, and the Thue–Morse arithmetic of Part II, are special to $b = 2$.

7.2 The finite 2-adic alias hierarchy

For $p > m$ the alias sum survives, but only through finitely many 2-adic shells.

Proposition 7.4 (Valuation hierarchy). *For every $p \in \mathbb{N}_0$,*

$$Q_{m,p}(\theta) - \int_{-1}^{1} x^p \operatorname{up}(x)\,\mathrm{d}x = \Big(-\frac{1}{2\pi\mathrm{i}}\Big)^p \sum_{v=0}^{p-m-1}\sum_{\substack{q\in\mathbb{Z}\\ q\text{ odd}}} \mathrm{e}^{2\pi\mathrm{i}2^v q\theta}\,\Phi^{(p)}\big(2^{m+v}q\big), \tag{7.4}$$

the outer sum being empty for $p \leq m$.

Proof. Write $\ell = 2^v q$ with q odd; the zero order at $M\ell$ is $m+v+1$, so $\Phi^{(p)}(M\ell) = 0$ unless $v \leq p-m-1$. □

Increasing the degree by one activates exactly one new shell. This finite stratification is what makes the defects *computable*: each shell is an odd-integer Dirichlet series built from a finite jet of the zero-stripped product, and the jets close under a Bell–Bernoulli calculus.

Proposition 7.5 (Bell–Bernoulli alias jets). *Let $n \neq 0$ be an integer, $d = \nu_2(n)+1$, $n = \sigma 2^{d-1}q$ with q odd, $\sigma = \pm 1$. Factoring the d vanishing sinc terms of* (1.8) *at n gives the exact local identity*

$$\Phi(n+w) = w^d\,\Psi_n(w), \qquad \Psi_n(w) = -\frac{P_d^{\mathrm{sinc}}(w)}{(n+w)^d}\,\Phi\Big(\frac{q}{2}+\frac{\sigma w}{2^d}\Big), \qquad P_d^{\mathrm{sinc}}(w) := \prod_{j=0}^{d-1}\operatorname{sinc}_\pi\Big(\frac{w}{2^j}\Big), \tag{7.5}$$

so Ψ_n is smooth and nonvanishing near $w = 0$ with $\Psi_n(0) = -\Phi(q/2)/n^d$. Let $\lambda_s(n) = (\log\Psi_n)^{(s)}(0)$. Then for every $r \geq 0$

$$\Phi^{(d+r)}(n) = \frac{(d+r)!}{r!}\,\Psi_n(0)\,Y_r\big(\lambda_1(n),\dots,\lambda_r(n)\big), \tag{7.6}$$

with

$$\lambda_s(n) = \bigl(\log P_d^{\mathrm{sinc}}\bigr)^{(s)}(0) + \frac{\sigma^s}{2^{ds}}\,\bigl(\log\Phi\bigr)^{(s)}\Bigl(\frac{q}{2}\Bigr) + (-1)^s\,\frac{d\,(s-1)!}{n^s}, \tag{7.7}$$

where the finite sinc block contributes nothing for odd s and, for $s = 2j$,

$$\bigl(\log P_d^{\mathrm{sinc}}\bigr)^{(2j)}(0) = -\frac{2^{2j-1}\,|B_{2j}|}{j}\,\pi^{2j}\,\frac{1-2^{-2jd}}{1-2^{-2j}}. \tag{7.8}$$

Proof. Equation (7.6) is the Leibniz expansion of $w^d\Psi_n(w)$ at $w = 0$ combined with the Bell formula $\Psi_n^{(r)}(0) = \Psi_n(0)Y_r(\lambda_1,\dots,\lambda_r)$ for derivatives of $\exp(\log\Psi_n)$. Logarithmic differentiation of the three factors of Ψ_n gives (7.7); for the finite sinc block use $\log\operatorname{sinc}_\pi z = -\sum_{j\ge1}\frac{2^{2j-1}|B_{2j}|}{j\,(2j)!}(\pi z)^{2j}$ and sum the geometric progression over the d scales. □

Together, (7.4) and (7.6) are a fully algorithmic exact description of $Q_{m,p}(\theta)$ for every natural degree: finitely many active shells, and every active derivative a finite Bell polynomial in explicit local data — Bernoulli logarithmic jets of the finite sinc block, logarithmic jets of Φ at half-integers, and inverse powers of n.

7.3 First failure and phase superconvergence

Theorem 7.6 (Degree-$(m{+}1)$ error)**.** *For $p = m+1$,*

$$Q_{m,m+1}(\theta) - \int_{-1}^{1} x^{m+1}\,\mathrm{up}(x)\,\mathrm{d}x = -\Bigl(-\frac{1}{2\pi\mathrm{i}}\Bigr)^{m+1}\frac{(m+1)!}{2^{m(m+1)}}\sum_{\substack{\ell\in\mathbb{Z}\\ \ell\ \mathrm{odd}}}\frac{\Phi(|\ell|/2)}{\ell^{m+1}}\,\mathrm{e}^{2\pi\mathrm{i}\ell\theta}. \tag{7.9}$$

As a trigonometric series in θ this is not identically zero (its first Fourier coefficient is a nonzero multiple of $\Phi(1/2)$), so the uniform degree m of theorem 7.1 is sharp.

Proof. Only the shell $v = 0$ survives in (7.4), and the leading jet (1.11) at the order-$(m{+}1)$ zero $M\ell = 2^m\ell$ gives $\Phi^{(m+1)}(M\ell) = -(m+1)!\,(M\ell)^{-(m+1)}\Phi(\ell/2)$. □

Corollary 7.7 (Phase superconvergence)**.** *At the first failing degree $p = m+1$:*

1. *if m is even, the centered and half-shifted combs are still exact: $Q_{m,m+1}(0) = Q_{m,m+1}(1/2) = 0$;*
2. *if m is odd, the quarter-shifted combs are still exact: $Q_{m,m+1}(1/4) = Q_{m,m+1}(3/4) = \int x^{m+1}\,\mathrm{up}$.*

Proof. For even m, $m+1$ is odd, and pairing $\ell \leftrightarrow -\ell$ in (7.9) produces $\sin(2\pi\ell\theta)$-type terms vanishing at $\theta \in \{0, \frac{1}{2}\}$ (the integral is also zero by symmetry). For odd m, the paired sum is proportional to $\cos(2\pi\ell\theta)$, which vanishes for every odd ℓ at $\theta \in \{\frac{1}{4}, \frac{3}{4}\}$. □

Exact rational experiments through $m = 4$ (companion script of the absorbed quadrature report, `assets/fabius_dyadic_comb_report_final/`) illustrate both the failure and the special phases:

m	$p = m+1$	shift θ	$Q_{m,p}(\theta) - \int x^p\,\mathrm{up}$
1	2	0	$1/72$
1	2	$1/4$	0
2	3	0	0
2	3	$1/4$	$-4643/11059200$
3	4	0	$-23/5529600$
3	4	$1/4$	0
4	5	0	0
4	5	$1/4$	$4966069/383551316951040$

Table 3: First-failure errors at the two distinguished phases; every tested shift was exact for all degrees $p \le m$.

Conjecture 7.8 (Phase classification)**.** *Modulo* 1*, the parity-forced zeros of theorem 7.7 are the only shifts that are superconvergent at degree* $m+1$ *for* every *level of the corresponding parity; individual levels may have accidental extra zeros.*

Remark 7.9 (Crosswalk: the first-failure series is a polynomial-synthesis defect). The trigonometric series (7.9) reappears, up to the explicit factor $(-1)^{m+1}2^{m(m+1)}$, as the Fourier residue of the canonical nonpolynomial defect of the local up-spline space in the companion volume *Exact Polynomial Synthesis from Rvachev Up-Atoms* (`representations/Up_Polynomial_Synthesis/`), whose defect–quadrature duality also converts the exact phase-error values of table 3 into the defect special values $\mathcal{E}_1(0) = \frac{1}{18}$, $\mathcal{E}_3(0) = -\frac{23}{1350}$ and, through the Dirichlet values (6.6), $\mathcal{E}_5(0) = \frac{1543}{119070}$, $\mathcal{E}_7(0) = -\frac{1196113}{65063250}$.

7.4 Absolute integer powers: the exact zeta correction

Write $U_{m,p} := U_m(p)$ for the punctured absolute comb (2.7) at integer degree. The Fabius collapse upgrades polynomial exactness to a closed form with a single correction term.

Theorem 7.10 (Absolute integer powers)**.** *For* $p \in \mathbb{N}_0$ *and* $m \ge \tau(p)$*,*

$$\boxed{U_{m,p} = \frac{2d_{p+1}}{p+1} + 2\zeta(-p)\,h^{p+1}.} \tag{7.10}$$

In particular: for positive even p *the comb equals the integral* (1.27) exactly*; for odd* p *the sole error is the rational correction* $2\zeta(-p)h^{p+1} = -B_{p+1}h^{p+1}\cdot\frac{2}{p+1}$*; and for* $p=0$ *the punctured comb is* $1-h$*.*

Proof. Let $p \ge 1$. By evenness and the fold (1.3), with $a = M-k$,

$$U_{m,p} = 2h\sum_{k=1}^{M}(kh)^p F(1-kh) = 2\sum_{j=0}^{p}(-1)^j\binom{p}{j}R_{m,j}$$

(the substitution produces the left-rule sums, which differ from the $R_{m,j}$ by h each; the correction $2h\sum_j(-1)^j\binom{p}{j} = 0$ vanishes precisely because $p \ge 1$). The same alternating binomial combination of the integrals I_j equals the absolute moment $2d_{p+1}/(p+1)$. Insert the exact formula (5.8) for each $R_{m,j}$ (all thresholds $\tau(j) \le \tau(p)$ are met). The alternating difference annihilates every Euler–Maclaurin term of degree $< p$ in h-power count; the only surviving boundary term arises at odd p and equals $2\zeta(-p)h^{p+1}$ by $\zeta(-p) = -B_{p+1}/(p+1)$. The case $p=0$ is the partition of unity minus the central sample $h\,\mathrm{up}(0) = h$. □

Once the threshold is reached, the first instances are exact identities:

$$U_{m,0} = 1-h,\quad U_{m,1} = \frac{5}{18} - \frac{h^2}{6},\quad U_{m,2} = \frac{1}{9},\quad U_{m,3} = \frac{143}{2700} + \frac{h^4}{60},\quad U_{m,4} = \frac{19}{675}. \tag{7.11}$$

Consistency bridge

The corpus's self-weighted shifted quadrature (samples of up used as weights) and the unweighted Fabius combs of section 5 are distinct constructions, but the elementary relation (2.8) makes them two faces of one mechanism: exact up-moments are *equivalent* to the cancellation between an ordinary Faulhaber sum and the exact Euler–Maclaurin polynomial of $L_{m,p}$. The zeta correction in (7.10) is precisely the odd-degree mismatch of the two boundary conventions, and its vanishing at even p is $\zeta(-2r) = 0$.

7.5 Finite arithmetic form

For exact rational work at fixed level, insert the convolution (1.29) directly: with $L = M - a - 1$,

$$U_{m,p} = \frac{2C_m}{M^{p+1}} \sum_{a=0}^{M-2} \varepsilon_a \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m}{2j} c_j \sum_{u=0}^{p} (-1)^u \binom{p}{u} L^{p-u} \sum_{v=0}^{m-2j} \binom{m-2j}{v} 2^v \frac{B_{u+v+1}(L) - B_{u+v+1}}{u+v+1}, \tag{7.12}$$

an identity of rational numbers for every m, p. When $m \geq \tau(p)$ it collapses to (7.10), producing a family of nontrivial finite Thue–Morse–Bernoulli identities; below threshold it evaluates the defect layers of theorem 7.4 exactly.

8 Complex, fractional, and negative powers

8.1 Entire dependence and the Hurwitz–Thue–Morse formula

At fixed level the comb $L_{m,\alpha}$ is a finite exponential sum, hence entire in α; and the continuous moment $I(\alpha) = (1 - d_{\alpha+1})/(\alpha+1)$ is entire by flatness (section 1.3). The finite comb has a closed form uniform in the exponent.

Theorem 8.1 (Hurwitz-zeta–Thue–Morse formula). *For every $m \in \mathbb{N}_0$ and $\alpha \in \mathbb{C}$,*

$$\begin{aligned} L_{m,\alpha} = \frac{C_m}{M^{\alpha+1}} \sum_{a=0}^{M-2} \varepsilon_a \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m}{2j} c_j \sum_{i=0}^{m-2j} \binom{m-2j}{i} 2^i (-2a-1)^{m-2j-i} \\ \times \Big[\zeta(-\alpha - i,\, a+1) - \zeta(-\alpha - i,\, M) \Big], \end{aligned} \tag{8.1}$$

each bracket being an entire function of α (the poles of the two Hurwitz zetas cancel). At $\alpha = p \in \mathbb{N}_0$ this reduces to theorem 4.1 via $\zeta(-n, x) = -B_{n+1}(x)/(n+1)$; at negative integers it becomes a finite combination of generalized harmonic numbers.

Proof. Insert (1.29) into (2.1), interchange the finite sums, and expand $(2k - 2a - 1)^{m-2j} = \sum_i \binom{m-2j}{i} (2k)^i (-2a-1)^{m-2j-i}$; the remaining finite power sum is $\sum_{k=a+1}^{M-1} k^{\alpha+i} = \zeta(-\alpha - i, a+1) - \zeta(-\alpha - i, M)$. □

Corollary 8.2 (Negative integer powers are rational). *For $q \in \mathbb{N}$,*

$$L_{m,-q} = M^{q-1} \sum_{k=1}^{M-1} \frac{F(k/M)}{k^q} \in \mathbb{Q}; \tag{8.2}$$

the case $q = 1$ is the harmonic Fabius comb $\sum_{k=1}^{M-1} F(k/M)/k$.

A second entire interpolation, numerically preferable near negative integers where the Hurwitz representation contains large cancelling terms, is the Newton form

$$L_{m,\alpha} = M^{-\alpha-1} \sum_{r=0}^{M-1} (\Delta^r 0^\alpha) \sum_{k=r}^{M-1} \binom{k}{r} F(k/M), \tag{8.3}$$

from $k^\alpha = \sum_{r \leq k} \binom{k}{r} \Delta^r 0^\alpha$. On the continuous side, expanding $F(x) = \mathrm{up}(1-x)$ around the flat point gives the rapidly convergent series

$$I(\alpha) = \sum_{r=0}^{\infty} (-1)^r \binom{\alpha}{r} a_r, \qquad a_r = \frac{d_{r+1}}{r+1} = \mathcal{O}_A(r^{-A}) \; \forall A, \tag{8.4}$$

locally uniformly in α: flatness of up at 1 makes the half-moments a_r decay faster than every power, so the generalized binomial series converges normally. For $\alpha = p \in \mathbb{N}_0$ it terminates and recovers (1.25).

8.2 Complex-power Euler–Maclaurin and Richardson filtering

Theorem 8.3 (Complex-power expansion). *For α in a compact $K \subset \mathbb{C}$ and every $R \geq 1$,*

$$L_{m,\alpha} = I(\alpha) - \frac{h}{2} + \sum_{r=1}^{R-1} \frac{B_{2r}}{(2r)!}\, \alpha^{\underline{2r-1}}\, h^{2r} + \mathcal{O}_{K,R}\big(h^{2R}\big) \qquad (m \to \infty). \tag{8.5}$$

Proof. The function $x^\alpha F(x)$, extended by 0 at $x = 0$, is C^∞ on $[0,1]$ uniformly for $\alpha \in K$: at 0 all derivatives vanish by flatness, and at 1 flatness of $1 - F$ gives $\big(x^\alpha F(x)\big)^{(n)}\big|_{x=1} = \alpha^{\underline{n}}$. The classical Euler–Maclaurin formula with R terms then has only the upper-endpoint corrections shown and a remainder $\mathcal{O}_{K,R}(h^{2R})$. □

When $\alpha = p \in \mathbb{N}_0$ the falling factorials terminate, and theorem 5.3 upgrades the truncated asymptotic to an exact identity above the threshold. For nonintegral α the series does not terminate, but it contains only even powers of h after the $h/2$ shift: the sequence $\widetilde{L}_m = L_{m,\alpha} + h/2$ Richardson-extrapolates with ratio 4, and for natural p the extrapolation *terminates in rational arithmetic*, returning I_p exactly once $m \geq \tau(p)$. Flatness also disarms the only apparent danger of negative exponents on the comb: the smallest node contributes

$$h^{1-s} F(h) = 2^{m(s-1)}\, C_m\, G_m(1), \tag{8.6}$$

which decays faster than every geometric sequence in m for fixed s, since $C_m = 2^{-m(m+1)/2}/m!$.

8.3 The Mellin transform of up

For $\Re\alpha > -1$ let $\mathcal{M}(\alpha) := \int_{-1}^{1} |x|^\alpha \operatorname{up}(x)\,\mathrm{d}x = 2\int_0^1 x^\alpha \operatorname{up}(x)\,\mathrm{d}x$. Since $\operatorname{up} - 1$ is flat at 0,

$$\mathcal{M}(\alpha) = \frac{2}{\alpha+1} + 2\int_0^1 x^\alpha \big(\operatorname{up}(x) - 1\big)\,\mathrm{d}x, \tag{8.7}$$

with an entire second term: $\mathcal{M}$ continues meromorphically to $\mathbb{C}$ with a *single* simple pole at $\alpha = -1$, of residue 2, and

$$\mathcal{M}(\alpha) = \frac{2d_{\alpha+1}}{\alpha+1} \tag{8.8}$$

identifies the continuation with the entire Fabius-law moment d_s. Thus d_s is simultaneously the analytic continuation of all absolute Rvachev moments — the meromorphic backbone of everything below.

8.4 The universal zeta correction

Lemma 8.4 (Cutoff-monomial summation). *Let $\chi \in C_c^\infty([0,\infty))$ equal 1 near 0, let $0 < \theta \leq 1$ and $\beta > -1$. Then for every $A > 0$*

$$h\sum_{k=0}^{\infty} \big(h(k+\theta)\big)^\beta \chi\big(h(k+\theta)\big) = \int_0^\infty x^\beta \chi(x)\,\mathrm{d}x + \zeta(-\beta,\theta)\,h^{\beta+1} + \mathcal{O}_{\beta,A,\chi}\big(h^A\big), \tag{8.9}$$

and the identity continues meromorphically in β with the integral read as a Mellin finite part; the two displayed residues collide only at $\beta = -1$.

Proof. Let $\chi^*(s) = \int_0^\infty \chi(x) x^{s-1}\,\mathrm{d}x$; integration by parts, $\chi^*(s) = -s^{-1}\int_0^\infty \chi'(x) x^s\,\mathrm{d}x$, shows χ^* is meromorphic with a single pole at $s = 0$ of residue 1 and rapid decay on vertical lines (as χ' has compact support in $(0,\infty)$). For $c > \beta + 1$, Mellin inversion gives

$$h^{\beta+1}\sum_{k\geq 0}(k+\theta)^\beta \chi\big(h(k+\theta)\big) = \frac{1}{2\pi\mathrm{i}}\int_{(c)} \chi^*(s)\,\zeta(s-\beta,\theta)\,h^{\beta+1-s}\,\mathrm{d}s.$$

Shift the contour to $\Re s = -A$: the Hurwitz pole at $s = \beta + 1$ contributes $\chi^*(\beta + 1) = \int x^\beta \chi$, the pole of χ^* at $s = 0$ contributes $\zeta(-\beta, \theta)h^{\beta+1}$, and polynomial vertical bounds for $\zeta(\cdot, \theta)$ against the rapid decay of χ^* bound the shifted integral by $\mathcal{O}(h^{\beta+1+A})$. □

Theorem 8.5 (Fractional absolute powers)**.** *Fix $\alpha > -1$. For every $A > 0$,*

$$\boxed{U_m(\alpha) = \mathcal{M}(\alpha) + 2\zeta(-\alpha)\, h^{\alpha+1} + \mathcal{O}_{\alpha,A}\big(h^A\big).} \tag{8.10}$$

For $\alpha = p \in \mathbb{N}_0$ the remainder vanishes identically once $m \geq \tau(p)$, by theorem 7.10.

Proof. Split $x^\alpha \operatorname{up}(x) = x^\alpha \chi(x) + x^\alpha(\operatorname{up}(x) - \chi(x))$ on $[0, \infty)$ with χ as in the lemma, supported inside $[0, 1)$. The second summand extends by zero to $C_c^\infty(0, \infty)$ — flatness of up -1 removes the origin, flatness of up the support endpoint — so Poisson summation makes its comb error $\mathcal{O}(h^A)$ for every A. Apply theorem 8.4 with $\theta = 1$ to the first summand and double by evenness. □

The striking feature is the *absence* of an infinite Euler–Maclaurin tail: flatness of up -1 kills every higher local coefficient, leaving one universal singular term. At integer exponents the trivial zeros $\zeta(-2r) = 0$ explain the even/odd parity of theorem 7.10. For a shifted comb the correction splits into its two one-sided lattice contributions:

Theorem 8.6 (Shifted Hurwitz corrections)**.** *For fixed $\alpha > -1$, $0 < \theta < 1$, and every $A > 0$,*

$$h \sum_{k\in\mathbb{Z}} |h(k+\theta)|^\alpha \operatorname{up}\big(h(k+\theta)\big) = \mathcal{M}(\alpha) + h^{\alpha+1}\big[\zeta(-\alpha,\theta) + \zeta(-\alpha, 1-\theta)\big] + \mathcal{O}(h^A), \tag{8.11}$$

$$h \sum_{k\in\mathbb{Z}} \operatorname{sgn}(k+\theta)\ |h(k+\theta)|^\alpha \operatorname{up}\big(h(k+\theta)\big) = h^{\alpha+1}\big[\zeta(-\alpha,\theta) - \zeta(-\alpha, 1-\theta)\big] + \mathcal{O}(h^A). \tag{8.12}$$

Proof. The shifted lattice has positive distances $h(k + \theta)$ and negative distances $h(k + 1 - \theta)$, $k \geq 0$; apply theorem 8.4 with shifts θ and $1 - \theta$ to the two singular halves, Poisson summation to the flat remainders, and add or subtract. The signed continuous integral vanishes by evenness. □

At natural exponents, the reflection identity for Bernoulli polynomials reconciles (8.11) with the exact shifted quadrature of theorem 7.1 and the phase zeros of theorem 7.7; a chosen complex branch on the negative axis is recovered from the absolute and signed combs via $x^\alpha = \frac{1+e^{\pi i\alpha}}{2}\,|x|^\alpha + \frac{1-e^{\pi i\alpha}}{2}\operatorname{sgn}(x)\,|x|^\alpha$.

8.5 Negative powers and the logarithmic transition

Proposition 8.7 (Renormalized negative powers)**.** *For fixed $\alpha < -1$,*

$$U_m(\alpha) - 2\zeta(-\alpha)h^{\alpha+1} \longrightarrow \mathcal{M}(\alpha) \qquad (m \to \infty), \tag{8.13}$$

with superalgebraic convergence, $\mathcal{M}(\alpha)$ denoting the meromorphic value (8.7). At the pole,

$$\boxed{U_m(-1) = 2\log M + 2\gamma + 2\int_0^1 \frac{\operatorname{up}(x) - 1}{x}\,dx + \mathcal{O}_A\big(h^A\big) \quad \forall A > 0.} \tag{8.14}$$

Proof. Away from $\alpha = -1$, theorem 8.4 in its meromorphically continued form gives (8.13). At $\alpha = -1 + \varepsilon$ the two singular objects have cancelling poles:

$$\mathcal{M}(-1+\varepsilon) = \frac{2}{\varepsilon} + 2\int_0^1 \frac{\operatorname{up}(x) - 1}{x}\,dx + \mathcal{O}(\varepsilon), \qquad 2\zeta(1-\varepsilon)h^\varepsilon = -\frac{2}{\varepsilon} + 2\log M + 2\gamma + \mathcal{O}(\varepsilon),$$

and the finite part at $\varepsilon = 0$ is (8.14). □

Remark 8.8 (Elementary route and the Fabius constant). Through the bridge (2.8), the critical case is also elementary: $U_m(-1) = 2(H_{M-1} - L_{m,-1})$ with H_n the harmonic number, and the complete Euler–Maclaurin tails of H_{M-1} and of $L_{m,-1}$ (theorem 8.3 at $\alpha = -1$, where $(-1)^{\underline{2r-1}} = -(2r-1)!$ makes the corrections $-\sum B_{2r}h^{2r}/(2r)$) cancel *term by term*, leaving

$$U_m(-1) = 2\log M + 2\big(\gamma - I(-1)\big) + \mathcal{O}_A(h^A), \tag{8.15}$$

consistent with (8.14) via the elementary identity $\int_0^1 \frac{\mathrm{up}(x)-1}{x}\,\mathrm{d}x = -I(-1)$, i.e. the constant is $\gamma - \mathbb{E}[-\log Y]$. For integer $q > 1$ the same cancellation gives the zeta-dominated law

$$U_m(-q) = 2\zeta(q)\,M^{q-1} - 2\left(\frac{1}{q-1} + I(-q)\right) + \mathcal{O}_A(h^A), \tag{8.16}$$

a discrete Mellin renormalization: the lattice singularity $2\zeta(q)M^{q-1}$ plus the finite part of the Rvachev integral. The only divergences of the whole family are the universal logarithm at $\alpha = -1$ and these zeta-driven powers below it; every Fabius comb $L_{m,\alpha}$, by contrast, converges for *every* complex α, since F is flat where the monomial is singular.

9 Iterated and fractional-order summation

9.1 The binomial kernel and two exact reductions

For a sequence $a = (a_k)$ let $(\Sigma a)_n = \sum_{k\le n} a_k$ and Σ^n its n-fold iterate. Induction with the hockey-stick identity gives the discrete Cauchy kernel

$$(\Sigma^n a)_K = \sum_{k\le K} \binom{K-k+n-1}{n-1} a_k. \tag{9.1}$$

Take $a_k = (k/M)^\alpha F(k/M)$ on $1 \le k \le M-1$ and define the normalized n-fold comb

$$L^{[n]}_{m,\alpha} := h^n\big(\Sigma^n a\big)_{M-1} = h^n \sum_{k=1}^{M-1} \binom{M-k+n-2}{n-1} \left(\frac{k}{M}\right)^\alpha F\left(\frac{k}{M}\right). \tag{9.2}$$

Two closed reductions to one-fold combs are available, one through Stirling numbers and one through elementary symmetric polynomials; they are equivalent, and each is more convenient in its own regime.

Theorem 9.1 (Exact n-fold reduction)**.** *For $n \ge 1$ and every $\alpha \in \mathbb{C}$,*

$$\boxed{L^{[n]}_{m,\alpha} = \frac{M^{1-n}}{(n-1)!} \sum_{j=0}^{n-1} {n-1 \brack j} M^j \sum_{i=0}^{j} (-1)^i \binom{j}{i} L_{m,\alpha+i}.} \tag{9.3}$$

Equivalently, with elementary symmetric polynomials e_r,

$$L^{[n]}_{m,\alpha} = \frac{1}{(n-1)!} \sum_{i=0}^{n-1} (-1)^i\, e_{n-1-i}\big(1,\, 1{+}h, \dots, 1{+}(n{-}2)h\big)\, L_{m,\alpha+i}. \tag{9.4}$$

An n-fold sum therefore requires only the n shifted base combs $L_{m,\alpha}, \dots, L_{m,\alpha+n-1}$.

Proof. The kernel is a polynomial of degree $n-1$ in the node: with $x = kh$,

$$h^{n-1}\binom{M-k+n-2}{n-1} = \frac{1}{(n-1)!}\prod_{s=0}^{n-2}\big(1 + sh - x\big),$$

since $(M-k-1+r)h = 1+(r-1)h-x$ for $r=1,\dots,n-1$. Expanding the product in powers of x through the elementary symmetric polynomials of $1, 1{+}h, \dots, 1{+}(n{-}2)h$, and recognizing $h\sum_k x^i (k/M)^\alpha F(k/M) = L_{m,\alpha+i}$, gives (9.4). Alternatively, expand the same product of consecutive terms as the rising factorial $\prod_{r=1}^{n-1}(M-k-1+r) = (M-k)^{\overline{n-1}} = \sum_j \left[{n-1 \atop j}\right](M-k)^j$ and then $(M-k)^j$ binomially; the identity $\sum_k k^i (k/M)^\alpha F(k/M) = M^{i+1} L_{m,\alpha+i}$ collects the powers of M into (9.3). ☐

Corollary 9.2 (Closed forms at natural degree). *For $p \in \mathbb{N}_0$ and $m \ge \tau(p+n-1)$, every shifted comb in (9.4) is given by the exact terminating polynomial (5.7); hence $L^{[n]}_{m,p}$ is an explicit polynomial in h with rational coefficients formed from $I_p, \dots, I_{p+n-1}$, Bernoulli numbers, and elementary symmetric polynomials. The corresponding full-support Rvachev sums close in the moment basis alone: for $m \ge p+n-1$,*

$$h^n \sum_{k=-M}^{M} \binom{M-k+n-1}{n-1} (kh)^p \,\mathrm{up}(kh) = \frac{1}{(n-1)!} \sum_{i=0}^{n-1} (-1)^i e_{n-1-i}\big(1{+}h, \dots, 1{+}(n{-}1)h\big)\, \mu_{p+i}, \tag{9.5}$$

with $\mu_{2r} = c_r$, $\mu_{2r+1} = 0$, by theorem 7.1 applied to the degree-$(p{+}n{-}1)$ polynomial weight.

The continuum limits are the classical Volterra kernels: for every $\alpha \in \mathbb{C}$,

$$\lim_{m\to\infty} L^{[n]}_{m,\alpha} = \frac{1}{(n-1)!}\int_0^1 (1-x)^{n-1} x^\alpha F(x)\,\mathrm{d}x = \frac{1}{(n-1)!}\sum_{i=0}^{n-1}(-1)^i \binom{n-1}{i} I(\alpha+i). \tag{9.6}$$

9.2 Fractional order

The generating-function form $(\Sigma^n a)_K = [z^K]\,(1-z)^{-n} A(z)$ continues canonically in the order: for $\Re\rho > 0$ define

$$L^{[\rho]}_{m,\alpha} := h^\rho \sum_{k=1}^{M-1} \frac{\Gamma(M-k+\rho-1)}{\Gamma(\rho)\,\Gamma(M-k)} \left(\frac{k}{M}\right)^\alpha F\!\left(\frac{k}{M}\right) = h^\rho\,[z^{M-1}]\,(1-z)^{-\rho} A_{m,\alpha}(z), \tag{9.7}$$

where $A_{m,\alpha}(z) = \sum_k (k/M)^\alpha F(k/M) z^k$. The discrete semigroup law $\Sigma^\rho\Sigma^\sigma = \Sigma^{\rho+\sigma}$ is immediate from $(1-z)^{-\rho}(1-z)^{-\sigma} = (1-z)^{-\rho-\sigma}$, and the fractional Volterra limit holds:

Theorem 9.3 (Fractional Volterra limit). *For $\Re\rho > 0$, locally uniformly in $\alpha \in \mathbb{C}$,*

$$\lim_{m\to\infty} L^{[\rho]}_{m,\alpha} = \frac{1}{\Gamma(\rho)}\int_0^1 (1-x)^{\rho-1} x^\alpha F(x)\,\mathrm{d}x. \tag{9.8}$$

Proof. Uniform Stirling asymptotics give $h^{\rho-1}\Gamma(M-k+\rho-1)/[\Gamma(\rho)\Gamma(M-k)] \to (1-x)^{\rho-1}/\Gamma(\rho)$ at $x = k/M$; flatness of F at 0 and $\Re\rho > 0$ at $x=1$ supply an integrable majorant after splitting off finitely many boundary nodes. ☐

For nonintegral ρ there is no finite Stirling reduction: the kernel is not polynomial. Two structured replacements are (i) a convergent Newton series in the shifted combs $L_{m,\alpha+i}$ with generalized binomial coefficients, and (ii) for the *singular* weights of section 8, a term-by-term application of theorem 8.4 to the Taylor expansion of the kernel at the singular point, which produces a finite string of corrections $\sum_{r\ge 0} \kappa_r\,\zeta(-\alpha-r,\theta)\,h^{\alpha+r+1}$ with κ_r the kernel's Taylor coefficients — more than one zeta term, precisely because the iterated kernel is not locally constant. Developing the two-variable object $(\rho,\alpha) \mapsto L^{[\rho]}_{m,\alpha}$ as a special function (Mellin–Barnes representation, joint asymptotics, arithmetic at rational ρ) is posed in section 12.

10 Arithmetic consequences

The exact formulas bound denominators. With the (nonsharp) level denominator

$$\Delta_m = m!\, 2^{m(m+1)/2} \,(2\lfloor m/2\rfloor + 1)!! \prod_{j=1}^{\lfloor m/2\rfloor} (2^{2j} - 1), \qquad \Delta_m\, F(k/M) \in \mathbb{Z}\ \forall k, \tag{10.1}$$

one gets immediately $\Delta_m M^{p+1} L_{m,p} \in \mathbb{Z}$ and, with $\Lambda_M = \operatorname{lcm}(1, \ldots, M-1)$, $\Delta_m \Lambda_M^q L_{m,-q} \in \mathbb{Z}$; the factorials and powers of M visible in (9.3) propagate the bounds to iterated sums. These are deliberately crude: the exact Euler–Maclaurin identity typically forces large cancellation between the moment denominators inside I_p and the Bernoulli powers of two, and above the threshold the true denominator of $L_{m,p}$ is governed by (5.8) alone.

Problem 10.1 (Sharp valuations)**.** *Determine $\nu_2(\operatorname{den} L_{m,p})$ and the odd-prime valuations $v_\ell(\operatorname{den} L_{m,p})$ uniformly in m, p (likewise for iterated sums and negative powers). Above the threshold this is the arithmetic of d_{p+1}, Bernoulli denominators (von Staudt–Clausen), and powers of two; below it, the Bell–Prouhet formula* (4.7) *confines the signed part to $p+2$ terms and may expose a tractable valuation filtration through the Mersenne factors $2^{2j} - 1$.*

11 Verification for Part I

Three fully independent exact routes to the same rational numbers were implemented in the absorbed sources and re-run for this consolidation:

1. direct summation of exact dyadic samples (theorem 1.4 or the bit recursion (1.31));
2. the exchanged Bernoulli–Thue–Morse formula (4.1);
3. the Bell–Prouhet collapse (4.7).

The absorbed packages verify $L^{\text{direct}}_{m,p} = L^{\text{Bernoulli}}_{m,p} = L^{\text{Bell}}_{m,p}$ over representative ranges, the exactness matrix $\{\mathcal{R}_{m,p} = 0\} \Leftrightarrow \{m \geq \tau(p)\}$ through level 8 (table 2), the iterated reduction (9.3) against direct n-fold summation for $m \leq 6$, $p \leq 4$, $n \leq 4$, the shifted first-failure table 3, and the observed stabilization thresholds

p	0	1	2	3	4	5	6	7	8	9	10
first exact m	0	2	2	4	4	6	6	8	8	10	10
$\tau(p)$	0	2	2	4	4	6	6	8	8	10	10

in exact agreement with theorem 6.4. For nonintegral exponents, high-precision scans confirm the universal zeta correction: the corrected sequences $U_m(\alpha) - 2\zeta(-\alpha)h^{\alpha+1}$ stabilize to 13–16 digits already by $m = 8$ (e.g. to 0.4913969866527990 at $\alpha = \frac{1}{2}$, 0.1707208835839732 at $\alpha = \frac{3}{2}$, 2.7852273785136552 at $\alpha = -\frac{1}{2}$), with no visible algebraic tail, exactly as theorem 8.5 predicts. For this volume the defect table and the exactness matrix were additionally reproduced by a fresh implementation (route 1 with the bit recursion), which is also the source of the values $D(6)$ and $D(8)$ in (6.6).

12 Frontier directions for Part I

Beyond the conjectures already stated (theorems 6.4 to 6.6 and 7.8 and theorems 6.7 and 10.1), the absorbed sources converge on the following programs.

Problem 12.1 (Row-polynomial geometry)**.** *Write $\mathscr{F}_m(z) = Q_m(z)/(1-z)$. Determine the exact reciprocal symmetry of Q_m (cf.* (3.9)*), the multiplicities of its cyclotomic factors, and whether its noncyclotomic* *zeros obey a limiting curve or annulus law. The collision of the two descriptions of Q_m — Thue–Morse mask times moment kernel versus the functional equation — suggests a finite spectral factorization.*

- *Valuation-selective extrapolation.* Degree p activates only the shells $\nu_2(\ell) \le p-m-1$ (theorem 7.4); combining levels $m, \dots, m+r$ with coefficients annihilating the first r shells should give *exact finite* filters rather than asymptotic Richardson extrapolation, with a natural q-binomial description at $q = 2^{-1}$.
- *Double scaling in negative powers.* For $q = q_m$ growing with m, the smallest node contributes $M^{q_m-1}F(1/M)$ with $F(2^{-m})$ governed by the corpus's Lambert-W_{-1} endpoint asymptotics; balancing it against later nodes should yield a phase diagram in q_m/m — a direct meeting point of the endpoint asymptotics and the comb calculus.
- *Beyond-all-orders remainders.* After the zeta correction, the fractional remainder in (8.10) is smaller than every power of h; its logarithm should be controlled by the same saddle geometry and Lambert-W scales as the Fourier decay of Φ. An explicit transseries, uniform in α, is the natural target.
- *Inverse-Fabius combs.* For $h\sum_k x_k^p F^{-1}(x_k)$ the integrand has logarithmic/Lambert endpoint behavior rather than flatness; singular Euler–Maclaurin should produce non-power corrections, and the exact inverse-dyadic values of the corpus may permit arithmetic special cases.
- *Nonintegral scale ratios.* For $\prod_j \operatorname{sinc}_\pi(q^j\xi)$ with $1/q \notin \mathbb{N}$, exact lattice alignment fails; seek finite multiscale filters (Gaussian-binomial coefficients are the natural candidate) whose symbols vanish to prescribed order at the approximate aliases.
- *Base-b and digital masks.* The Bell–Prouhet mechanism needs only a digital mask with a high-order zero at $z = 1$; classify the masks that also create high-order zeros at the dual lattice, and hence an exact Euler–Maclaurin range, as in theorem 7.3.

Formalization roadmap for Part I

The most modular Lean targets, in dependency order: the convolution identity (1.29) and row OGF with pole cancellation (finite algebra); the finite-product identity (4.5) and the Bell moments (4.6); the Bernoulli sum exchange (4.1); a periodic trapezoidal alias theorem for a function with one endpoint jump; the collapse theorem 5.3 from the already-formalized integer zero orders of Φ; the Stirling reduction (9.3); and only then the analytic Hurwitz/fractional layers. The collapse is especially suitable: after the zero-order lemma, every forbidden alias derivative is literally zero, so the essential spectral argument is finite.

Part II

Global polynomial interpolation on the dyadic comb

13 The interpolation problem and its classical context

Part I used the dyadic samples as *sums*; Part II asks what happens when a single global polynomial is required to pass *through* them. The Fabius function is an unusually clean test case for this question: the data are arithmetically ideal (exact rationals, by section 1.4), the target is C^∞ with exact symmetry, and it is infinitely flat at both endpoints — yet it is nowhere analytic, so every analyticity-based convergence mechanism is unavailable. The result is a particularly sharp laboratory for Runge instability, in which data accuracy, operator conditioning, and evaluation stability can be completely separated.

Three classical facts frame the study. For $M+1$ equidistant nodes the Lebesgue constant of ordinary interpolation grows exponentially,

$$\Lambda_M^{\mathrm{eq}} \sim \frac{2^{M+1}}{\mathrm{e}\, M \log M} \tag{13.1}$$

[19, 20, 21]; barycentric evaluation [22, 23] stabilizes the *evaluation* of a given interpolant but cannot change this operator norm; and no method using equispaced samples can combine stability with unrestricted exponential convergence (Platte–Trefethen–Kuijlaars [25]; see also Coppersmith–Rivlin [24] for how a polynomial bounded on the grid can grow like 2^M between nodes). For the Fabius data the classical pointwise remainder is formally exact but useless: with $\|F^{(M+1)}\|_\infty = 2^{\binom{M+2}{2}}$ from (1.7), the factor $2^{(M+1)(M+2)/2}/(M+1)!$ overwhelms every nodal-product bound. The right diagnostics are instead the *exact cardinal geometry* of the comb (section 16) and the arithmetic of the exact data (section 19).

Within the corpus, two neighboring bodies of work are deliberately *not* duplicated here: the geometric-node (q-binomial) Lagrange formulas used as coefficient extractors, whose nodes are multiplicative rather than the additive comb j/M; and the stable local dyadic reconstructions (piecewise-linear/Faber–Schauder, repeated-integration splines). The audited corpus contained no global additive-comb Runge study and no endpoint-flat confluent construction before the six absorbed reports; their independent audits agree on this boundary (Appendix E).

14 Ordinary global Lagrange interpolation

Let $\mathcal{I}_M f$ denote the unique polynomial of degree at most M with $(\mathcal{I}_M f)(x_j) = f(x_j)$ at the comb nodes $x_j = j/M$, $0 \le j \le M$, and write $P_M := \mathcal{I}_M F$.

14.1 Barycentric and Newton forms

For equispaced nodes the barycentric weights are $\lambda_j = (-1)^j \binom{M}{j}$:

$$P_M(x) = \frac{\displaystyle\sum_{j=0}^{M} \frac{(-1)^j \binom{M}{j}}{x - x_j} F(x_j)}{\displaystyle\sum_{j=0}^{M} \frac{(-1)^j \binom{M}{j}}{x - x_j}} \qquad (x \notin \{x_j\}), \tag{14.1}$$

and the Newton-forward form is

$$P_M(x) = \sum_{k=0}^{M} \binom{Mx}{k} \Delta^k F(0), \qquad \Delta^k F(0) = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} F(j/M). \tag{14.2}$$

The ordinary Pascal row here is a $q = 1$ object sitting on top of $q = \frac{1}{2}$ data — the mixed transform theme developed in section 19.

14.2 Symmetry, degree drop, and compressed forms

Proposition 14.1 (Symmetry and exact degree drop)**.** *For even M, P_M has rational coefficients,*

$$P_M(1-x) = 1 - P_M(x), \tag{14.3}$$

and $\deg P_M \le M-1$*; equivalently the top forward difference vanishes,* $\Delta^M F(0) = 0$*. Consequently there is a rational polynomial* R_M *with*

$$P_M(x) = x + x(1-x)(2x-1)\, R_M\big(x(1-x)\big), \qquad \deg R_M \le \frac{M}{2} - 2, \tag{14.4}$$

and the interpolant of up *on the symmetric comb* $t_j = -1 + 2j/M$ *of* $[-1,1]$ *is even:* $\mathcal{I}_M \,\mathrm{up}(t) = (1-t^2)\, T_M(t^2)$ *with* $\deg T_M \le M/2 - 1$.

Proof. The polynomial $1 - P_M(1-x)$ matches the data by (1.5), so uniqueness gives (14.3). Centering at $x = \frac{1}{2}$, the function $P_M(\frac{1}{2} + y) - \frac{1}{2}$ is odd in y; its a priori degree bound M is even, so the leading coefficient vanishes. Then $P_M(x) - x$ is antisymmetric about $\frac{1}{2}$ and vanishes at $0, \frac{1}{2}, 1$; dividing by $x(1-x)(2x-1)$ leaves a reflection-symmetric polynomial, i.e. a polynomial in $x(1-x)$. The up statement is the same argument with evenness and the zeros at $t = \pm 1$. □

The compression halves the working dimension and provides exact symmetry checks for code; it does *not* improve conditioning — the compressed coefficients are as enormous as the original ones.

14.3 A 2-adic law for the barycentric weights

The real magnitudes of the weights drive the instability below, but on the dyadic comb the same weights are arithmetically stratified.

Theorem 14.2 (2-adic valuation of the comb weights). *For* $M = 2^m$ *and* $1 \le j \le M-1$,

$$\nu_2 \binom{M}{j} = m - \nu_2(j). \tag{14.5}$$

For the interior weights $\binom{M-2}{j-1}$ *of section 15, Lucas's theorem gives oddness exactly for odd* j*, with higher valuations determined by the binary carries of* $j-1$ *against* $M-2$.

Proof. $\binom{M}{j} = \frac{M}{j}\binom{M-1}{j-1}$, and every entry of row $M - 1 = 2^m - 1$ is odd by Lucas's theorem, since all binary digits of $M-1$ are 1. □

Thus the weights that are catastrophic in the real norm are perfectly regular 2-adically — a hint (made precise in theorem 21.5) that a renormalized 2-adic interpolation problem on the same data may be far better behaved than the archimedean one.

14.4 Endpoint-jet defects: the Stirling transform

Differentiating (14.2) at 0 with $(Mx)^{\underline{k}} = \sum_r s(k,r) M^r x^r$ ($s(k,r)$ the signed Stirling numbers):

Proposition 14.3 (Exact endpoint jets). *For* $0 \le r \le M$,

$$(P_M)^{(r)}(0) = r!\, M^r \sum_{k=r}^{M} \frac{s(k,r)}{k!}\, \Delta^k F(0); \tag{14.6}$$

in particular the endpoint slope is the mixed binomial transform

$$(P_M)'(0) = M \sum_{j=1}^{M} (-1)^{j-1} \binom{M}{j} \frac{F(j/M)}{j}. \tag{14.7}$$

Since $F^{(r)}(0) = 0$ for $r \ge 1$, every value (14.6) is a *purely interpolatory* boundary defect, computable as an exact rational number. Constraining the first r endpoint derivatives (section 15) removes exactly the first r defects; this explains why low-order constraints help at finite M, and says nothing about the remaining global modes — the subject of section 16.

14.5 The observed Runge transition

High-precision scans with exact rational data (protocol in section 18) show the classical Runge geometry with a sharp onset:

M	$\|F-P_M\|_\infty$ (sampled)	leftmost maximizer	sampled Λ_M	$\log_2\|F-P_M\|_\infty$
4	3.997×10^{-2}	0.0923	2.21	-4.6
8	9.487×10^{-3}	0.0406	1.10×10^{1}	-6.7
16	3.797×10^{-3}	0.0154	9.35×10^{2}	-8.0
32	4.989	0.0071	2.43×10^{7}	2.3
64	3.589×10^{7}	0.0031	4.41×10^{16}	25.1
128	3.837×10^{22}	0.0013	3.54×10^{35}	75.0

Table 4: Ordinary interpolation of F on the comb j/M. Errors are sampled maxima on dyadic grids of mesh 2^{-12}–2^{-13}; the Lebesgue constants are sampled on the same grids. The maximizer migrates toward the endpoint on the scale $1/M$.

The error *decreases* through $M=16$ — a misleading impression of convergence — then explodes in narrow boundary layers. Because samples and comparison values are exact rationals and the precision was raised until every displayed digit stabilized, this is a property of the interpolant, not of the arithmetic.

A still cleaner observable is the exact endpoint derivative (14.7), whose true value should be 0:

M	sign	$\log_2\|(P_M)'(0)\|$	$M-\log_2\|(P_M)'(0)\|$
16	+	-0.681	16.681
32	−	10.816	21.184
64	+	34.823	29.177
128	−	85.942	42.058
256	−	206.287	49.713
512	−	450.491	61.509
1024	+	949.989	74.011
2048	+	1957.083	90.917
4096	−	3992.945	103.055
8192	−	8070.574	121.426

Table 5: The exact rational endpoint derivative defect of the ordinary interpolant, computed from (14.7) before taking logarithms. The binary deficit in the last column is consistent with a quadratic function of $\log_2 M$ plus lower-order oscillation; the sign pattern is irregular.

Numerical observation 14.4. The interpolants P_{2^m} exhibit the classical endpoint oscillation of Runge's phenomenon, with $\log_2\|F-P_M\|_\infty\approx M-o(M)$ at the tested sizes and an exact endpoint derivative growing almost like 2^M.

Conjecture 14.5 (Fabius Runge divergence). *$\|F-P_{2^m}\|_\infty\to\infty$; more strongly, $\frac{1}{M}\log_2\|F-P_M\|_\infty\to 1$ along dyadic M, possibly with a subexponential, dyadically oscillatory correction.*

Conjecture 14.6 (Endpoint derivative law). *There are constants $c_1,c_2>0$ with*

$$M-c_2(\log_2 M)^2\le\log_2\left|(P_M)'(0)\right|\le M-c_1(\log_2 M)^2 \tag{14.8}$$

for all large dyadic M outside a possible sparse zero set, the remainder after the quadratic-log term containing a bounded or slowly growing log-periodic component tied to the endpoint phase of F.

The operator theory of section 16 makes divergence plausible but does not prove it for this specific data vector: that requires controlling the cancellation between exponentially large cardinal terms and the special Thue–Morse structure of the exact values — the central open problem of Part II (section 21). A proof of theorem 14.6 would already imply nonuniform convergence, and with a quantitative Markov inequality, a large supremum norm in a boundary interval.

15 The endpoint-flat Hermite family

The most natural use of the flatness (1.6) is to impose the *true* endpoint derivatives — all zero — on the interpolant. The resulting confluent problem has an exact factorization that converts it into ordinary interior interpolation of a deflated quotient, exposes the entire conditioning geometry, and contains ordinary Lagrange interpolation as its zeroth member.

15.1 The smoothstep and the factorization

For $r \geq 0$ define the symmetric beta-polynomial smoothstep and the endpoint weight

$$S_r(x) := I_x(r+1, r+1) = \frac{(2r+1)!}{(r!)^2} \int_0^x t^r (1-t)^r \, \mathrm{d}t = \frac{(2r+1)!}{(r!)^2} \sum_{k=0}^{r} (-1)^k \binom{r}{k} \frac{x^{r+k+1}}{r+k+1}, \tag{15.1}$$

$$W_r(x) := \big[x(1-x)\big]^{r+1}. \tag{15.2}$$

S_r is the unique polynomial of degree $2r+1$ with $S_r(0) = 0$, $S_r(1) = 1$, vanishing derivatives through order r at both endpoints, and $S_r(1-x) = 1 - S_r(x)$; the first members are $S_0 = x$, $S_1 = 3x^2 - 2x^3$, $S_2 = 10x^3 - 15x^4 + 6x^5$.

Because $F - S_r$ has zeros of order at least $r + 1$ at both endpoints, the deflated quotient

$$g_r(x) := \frac{F(x) - S_r(x)}{W_r(x)} \tag{15.3}$$

extends to a C^∞ function on $[0, 1]$ with $g_r(1-x) = -g_r(x)$. Let $\mathcal{I}^\circ_{M-2}$ denote ordinary interpolation of degree at most $M - 2$ on the *interior* comb $x_j = j/M$, $1 \leq j \leq M - 1$.

Theorem 15.1 (Endpoint-flat factorization). *For $M = 2^m \geq 4$ and $r \geq 0$, define*

$$H_{M,r} := S_r + W_r \cdot \mathcal{I}^\circ_{M-2}\, g_r. \tag{15.4}$$

Then $H_{M,r}$ is the unique polynomial of degree at most $M + 2r$ satisfying

$$H_{M,r}(j/M) = F(j/M) \quad (0 \leq j \leq M), \qquad H^{(q)}_{M,r}(0) = H^{(q)}_{M,r}(1) = 0 \quad (1 \leq q \leq r), \tag{15.5}$$

and it satisfies the exact error identity and symmetry

$$F - H_{M,r} = W_r \cdot \big(g_r - \mathcal{I}^\circ_{M-2} g_r\big), \qquad H_{M,r}(1-x) = 1 - H_{M,r}(x). \tag{15.6}$$

All coefficients are rational; the actual degree is at most $M + 2r - 1$; and $H_{M,0} = P_M$. Explicitly, the interior data are

$$\mathcal{I}^\circ_{M-2} g_r \text{ interpolates } \quad g_r(j/M) = \frac{F(j/M) - S_r(j/M)}{\big[\frac{j}{M}(1 - \frac{j}{M})\big]^{r+1}}, \qquad 1 \leq j \leq M - 1, \tag{15.7}$$

and the compressed form $H_{M,r}(x) = S_r(x) + W_r(x)(2x-1)\, R_{M,r}\big(x(1-x)\big)$ holds with $\deg R_{M,r} \leq M/2 - 2$.

Proof. The second term of (15.4) has degree at most $(M-2) + (2r+2) = M + 2r$. At an interior node the definition of g_r gives the value conditions; at 0 and 1 the factor W_r vanishes to order $r+1$, so values and derivatives through order r are those of S_r, which are correct. For uniqueness, the homogeneous problem asks for a polynomial of degree at most $M+2r$ divisible by $x^{r+1}(1-x)^{r+1} \prod_{j=1}^{M-1}(x - j/M)$, of degree $M + 2r + 1$; only the zero polynomial qualifies. Subtracting (15.4) from $F = S_r + W_r g_r$ gives the error identity. Symmetry follows by applying uniqueness to $1 - H_{M,r}(1-x)$; the odd central part then kills the top even coefficient, and dividing the antisymmetric interior interpolant by $2x - 1$ produces the compressed form as in theorem 14.1. For $r = 0$: $S_0 = x$, $W_0 = x(1-x)$, and $H_{M,0}$ matches all $M + 1$ Lagrange conditions with degree $\leq M$. □

Remark 15.2 (Rvachev versions). Two distinct up problems reduce to the same machinery. *(i) Half comb.* On $[-1, 0]$, $\mathrm{up}(t) = F(1 + t)$, so interpolation on $t_j = -1 + j/M$ is the Fabius problem verbatim. For the symmetric comb $t_j = -1 + 2j/M$ of $[-1, 1]$, both endpoint values vanish, no smoothstep is needed, and

$$H^{\mathrm{up}}_{M,r}(t) = (1 - t^2)^{\,r+1}\, T_{M,r}(t^2), \qquad \deg T_{M,r} \le \tfrac{M}{2} - 1, \tag{15.8}$$

with $T_{M,r}$ interpolating $\mathrm{up}(t_j)/(1 - t_j^2)^{r+1}$; evenness is forced by uniqueness. *(ii) Full integer comb.* For the comb $y_j = j/M$, $-M \le j \le M$, of $[-1, 1]$, the degree-$2M$ interpolant R_M of up is even, so $R_M(y) = Q_M(y^2)$ where Q_M interpolates the data $\mathrm{up}(j/M)$ at the *quadratically spaced* nodes j^2/M^2 — an exact reduction of the symmetric equispaced problem to a one-sided problem on a quadratic mesh, useful both computationally and conceptually (the quadratic mesh clusters near 0, not near the dangerous support endpoints). The endpoint-flat versions carry the factor $(1 - y^2)^{r+1}$ exactly as in (i).

15.2 Barycentric implementation

On the interior comb the scaled barycentric weights are

$$w_j = (-1)^{j-1}\binom{M-2}{j-1}, \qquad 1 \le j \le M - 1, \tag{15.9}$$

and

$$\mathcal{I}^{\circ}_{M-2} g_r(x) = \frac{\displaystyle\sum_{j=1}^{M-1} \frac{w_j\, g_r(x_j)}{x - x_j}}{\displaystyle\sum_{j=1}^{M-1} \frac{w_j}{x - x_j}}. \tag{15.10}$$

No confluent Vandermonde system is ever formed: the exact rational data (15.7) feed a standard second-form barycentric evaluation [22]. High precision remains essential — the formula avoids avoidable roundoff but cannot repair the exponentially ill-conditioned dependence on the $g_r(x_j)$ quantified next.

15.3 Why the textbook remainder explains nothing

The confluent Hermite remainder is formally exact: with $K = M + 2r + 1$, for each x there is ξ_x with

$$F(x) - H_{M,r}(x) = \frac{F^{(K)}(\xi_x)}{K!}\, x^{r+1}(x-1)^{r+1} \prod_{j=1}^{M-1}\Big(x - \frac{j}{M}\Big). \tag{15.11}$$

Inserting the sharp norms (1.7) produces the factor $2^{K(K+1)/2}/K!$ — astronomically large, and equally so in the regimes where the actual error is 10^{-6}. The failure of (15.11) to be predictive is another signature of nowhere analyticity: one extreme global derivative norm ignores all cancellation. The informative decomposition is (15.6) together with the cardinal geometry of the next section, and, for the actual Fabius data, the arithmetic cancellation phenomena of section 19.

16 Weighted Lebesgue geometry and the instability theorems

16.1 The value-data operator and its cardinal functions

Hold the endpoint values and jets exact and perturb the interior values by δ_j. By (15.4) the interpolant moves by

$$\delta H(x) = W_r(x) \sum_{j=1}^{M-1} \frac{\ell^{\circ}_j(x)}{W_r(x_j)}\, \delta_j, \tag{16.1}$$

where ℓ_j° are the interior cardinal polynomials. The relevant operator norm from ℓ^∞ data is therefore the *weighted Lebesgue function and constant*

$$\mathscr{L}_{M,r}(x) = \sum_{j=1}^{M-1} |L_{j,r}(x)|\,, \qquad L_{j,r}(x) := \left[\frac{x(1-x)}{x_j(1-x_j)}\right]^{r+1} \ell_j^\circ(x), \qquad \mathscr{L}_{M,r} = \max_{[0,1]} \mathscr{L}_{M,r}(x). \tag{16.2}$$

This is a lower bound on the conditioning of *any* implementation of the same polynomial operator, independent of the evaluation strategy; and it is invariant under the affine change to $[-1,1]$, so all statements below hold verbatim for the Rvachev versions of theorem 15.2. Note $r = 0$ recovers (the interior part of) the ordinary Lebesgue constant, and the weight ratio in $L_{j,r}$ is the two-sided competition in miniature: $W_r(x)$ *suppresses* output near the endpoints while $W_r(x_j)^{-1}$ *amplifies* data from nodes near them.

Everything is computable through one gamma-product formula.

Lemma 16.1 (Gamma form of the interior cardinals). *Let $y = Mx \in (0, M)$ be a non-node. Then, for $1 \le j \le M-1$,*

$$\left|\ell_j^\circ(x)\right| = \frac{\Gamma(y)\,\Gamma(M-y)\ |\sin \pi y|}{\pi\ |y-j|\ \Gamma(j)\,\Gamma(M-j)}. \tag{16.3}$$

Proof. With $\omega(y) = \prod_{k=1}^{M-1}(y-k)$, reflection gives $|\omega(y)| = \Gamma(y)\Gamma(M-y)\,|\sin \pi y|\,/\pi$, and $|\omega'(j)| = (j-1)!\,(M-1-j)! = \Gamma(j)\Gamma(M-j)$; divide. □

16.2 Exact modes

Three explicit evaluations of single terms of (16.2) organize the entire theory. Write them as *modes*, each an exact lower bound for $\mathscr{L}_{M,r}$.

Proposition 16.2 (Exact cardinal modes). *Let M be even.*

1. Boundary-to-center mode *($x = \frac{1}{2M}$, $j = \frac{M}{2}$):*

$$\mathscr{L}_{M,r} \ge A_{M,r} := A_M\, a_M^{r+1}, \qquad A_M = \frac{\Gamma(M-\frac12)}{\sqrt{\pi}\,(\frac{M}{2}-\frac12)\,\Gamma(M/2)^2}, \qquad a_M = \frac{2M-1}{M^2}, \tag{16.4}$$

with $A_M \sim 2^M/(\pi\sqrt{2}\,M)$; hence for every fixed r

$$A_{M,r} \sim \frac{2^{\,M+r+1}}{\pi\sqrt{2}\ M^{\,r+2}}. \tag{16.5}$$

2. Center-to-boundary mode *($x = \frac12 + \frac{1}{2M}$, $j = 1$):*

$$\mathscr{L}_{M,r} \ge B_{M,r} := B_M\, b_M^{r+1}, \qquad B_M = \frac{\Gamma(\frac{M}{2}-\frac12)^2}{\pi\,\Gamma(M-1)} \sim 2^{\,2-M}\sqrt{\frac{2}{\pi M}}, \qquad b_M = \frac{M+1}{4}. \tag{16.6}$$

3. Interior entropy modes*: for any fixed $a \in (0, \frac12)$, with $\mathsf{H}(a) = -a\log a - (1-a)\log(1-a)$,*

$$\frac{1}{M}\log\left|\ell_{M/2}^\circ(a)\right| \longrightarrow \log 2 - \mathsf{H}(a) > 0, \qquad \frac{W_r(a)}{W_r(1/2)} = \big[4a(1-a)\big]^{\,r+1}. \tag{16.7}$$

At arithmetically clean points the entropy modes are exact gamma products; e.g. at $x = \frac13$ (never a node, since $3 \nmid M$),

$$\left|L_{M/2,r}(\tfrac13)\right| = \left(\frac{8}{9}\right)^{r+1} \frac{3\sqrt{3}}{\pi M}\,\frac{\Gamma(M/3)\,\Gamma(2M/3)}{\Gamma(M/2)^2} = \Theta\Big(M^{-1}\mathrm{e}^{\delta M}(8/9)^r\Big), \tag{16.8}$$

where $\delta = \log 2 - \mathsf{H}(\frac13) = \frac53\log 2 - \log 3 = 0.0566330\ldots$.

Proof. All parts are theorem 16.1 at the stated points plus the elementary weight ratios; the gamma-product manipulations (duplication for B_M, reflection at $y = M/3$ for (16.8), Stirling for the asymptotics) are collected in Appendix A. □

The three modes have transparent meanings. Mode (i) measures how a perturbation of the *central* value contaminates the first half-cell: exponentially at $r = 0$, damped by only one factor $\sim 2/M$ per extra jet order. Mode (ii) measures how a perturbation at the *first interior node* contaminates the center: negligible at $r = 0$ but amplified by $(M + 1)/4$ per jet order, because the weight $W_r(x_1)^{-1}$ explodes. The entropy modes fill in the whole interior profile. The instability theory is simply the statement that these mechanisms cannot all be small at once.

16.3 Fixed order and all orders

Theorem 16.3 (Every fixed jet order is exponentially unstable)**.** *For every fixed $r \geq 0$, $\mathscr{L}_{M,r} \geq A_{M,r} = (1 + o(1))\, 2^{M+r+1}\big(\pi\sqrt{2}\, M^{r+2}\big)^{-1}$ grows exponentially in M. Each additional endpoint derivative buys only one algebraic factor $\asymp M^{-1}$ against 2^M.*

Theorem 16.4 (No jet schedule stabilizes the family)**.** *There exist absolute constants $c, C > 0$ and M_0 such that for every dyadic $M \geq M_0$ and* every *integer $r \geq 0$,*

$$\boxed{\mathscr{L}_{M,r} \geq C\, \mathrm{e}^{cM}.} \tag{16.9}$$

Consequently no sequence $r = r(M)$ of endpoint orders makes the endpoint-flat interpolation operators subexponentially conditioned on the dyadic comb.

Proof. Fix $a \in (0, \frac{1}{2})$, say $a = \frac{1}{4}$, and put $\alpha = \log 2 - \mathsf{H}(a) > 0$ and $\eta = \alpha/\big(2 \log \frac{1}{4a(1-a)}\big) > 0$ (for $a = \frac{1}{4}$, $4a(1 - a) = \frac{3}{4}$).

Moderate orders, $r \leq \eta M$. Keep the single entropy-mode term $j = M/2$ at $x = a$ (perturbing a by $\mathcal{O}(1/M)$ to avoid nodes changes nothing at exponential scale). Its logarithm is

$$M\big(\alpha + o(1)\big) + (r + 1) \log\big[4a(1 - a)\big] \geq M\alpha - \eta M \log \tfrac{1}{4a(1-a)} + o(M) = \tfrac{\alpha}{2} M + o(M),$$

so the mode is at least $\mathrm{e}^{c_1 M}$ for large M.

Large orders, $r > \eta M$. Use mode (ii):

$$\log B_{M,r} = (r + 1) \log \frac{M + 1}{4} - M \log 2 + \mathcal{O}(\log M) \geq \eta M \log \frac{M + 1}{4} - M \log 2 + \mathcal{O}(\log M),$$

which is eventually of order $M \log M$, in particular $\geq c_2 M$. Take $c = \min(c_1, c_2)$. □

Remark 16.5 (Interpretation, and the two regimes). Raising r first damps the boundary-to-center modes (each entropy mode carries $[4a(1 - a)]^{r+1} < 1$), but the division by $W_r(x_j)$ in (15.7) eventually makes data at near-endpoint nodes explosively influential in the interior — mode (ii). Endpoint flatness is a *preconditioner with two opposing regimes*, not a stability cure. The proof deliberately uses only single cardinal terms; the exact identities (16.4)–(16.8) make each ingredient a finite gamma-product statement, which is also what makes the theorem a realistic formalization target (section 22).

16.4 Matching derivatives at every node is worse

The other reading of "use derivative data" — prescribe $F^{(s)}(x_j)$ for $0 \leq s \leq q$ at *every* node — is exactly computable too, since the derivative tower (1.7) makes every datum $F^{(s)}(j/M) = 2^{\binom{s+1}{2}} \mathcal{F}(2^s j/M)$ an explicit rational number. It fails faster.

Theorem 16.6 (First-order global Hermite amplification). *Let $\mathcal{H}_M$ interpolate F and F' at all $M+1$ comb nodes (degree $\le 2M+1$), and let ℓ_j be the full-comb cardinals. The cardinal multiplying a value perturbation at x_j is $h_j = [1 - 2\ell_j'(x_j)(x - x_j)]\,\ell_j^2$; at the central node $\ell'_{M/2}(x_{M/2}) = 0$ by symmetry, and at $x = \frac{1}{2M}$*

$$\left|\ell_{M/2}(\tfrac{1}{2M})\right| = \frac{\Gamma(M+\frac{1}{2})}{2\sqrt{\pi}\,(\frac{M}{2}-\frac{1}{2})\,\Gamma(\frac{M}{2}+1)^2} \sim \frac{\sqrt{2}}{\pi}\frac{2^M}{M^2}, \tag{16.10}$$

so the value-data Lebesgue constant of the scheme obeys

$$\Lambda_M^{C^1} \ge \left|h_{M/2}(\tfrac{1}{2M})\right| \sim \frac{2}{\pi^2}\frac{4^M}{M^4}. \tag{16.11}$$

Proof. $\ell'_{M/2}(x_{M/2}) = M\sum_{k\ne M/2}(M/2-k)^{-1} = 0$ by symmetry of the full comb; the gamma product is theorem 16.1 adapted to the full node set (Appendix A); and $h_{M/2} = \ell^2_{M/2}$ at this point. □

Squaring the cardinals squares the exponential. The numerical confirmation is immediate and extends to higher confluence:

M	q (derivatives matched)	degree	sampled max error
4	1	9	1.148×10^{-2}
4	2	14	6.555×10^{-3}
4	3	19	6.887×10^{-3}
8	1	17	3.736×10^{-3}
8	2	26	4.435×10^{-3}
8	3	35	2.735×10^{-1}
16	1	33	2.272
16	2	50	5.089×10^{2}
32	1	65	1.538×10^{7}
64	1	129	1.352×10^{23}

Table 6: Full-comb confluent Hermite interpolation with exact rational values *and* derivatives. The explosion arrives at half the size of the ordinary case (table 4), consistent with the squared cardinal geometry.

Conjecture 16.7 (Higher confluent amplification). *For fixed $s \ge 1$, matching derivatives through order s at all nodes gives $\Lambda^{\text{confl}}_{M,s} \ge C_s\, 2^{(s+1)M} M^{-\alpha_s}$ for an explicit α_s; theorem 16.6 is $s = 1$ with $\alpha_1 = 4$.*

The moral is not that derivative information is useless — section 20 shows local Hermite data succeeding completely — but that *where* confluent conditions are imposed matters more than their exactness: piling them onto one global equispaced polynomial inherits, and can square, the same cardinal geometry.

17 The balance law and the logarithmically thin window

The operator is exponentially unstable for every jet order — and yet the exact Fabius data exhibit a narrow range of r in which the actual error drops by many orders of magnitude (section 18). The location of that window is predicted almost perfectly by balancing the two exact modes of theorem 16.2.

17.1 Exact and asymptotic balance orders

Definition 17.1 (Two-mode balance order). For even $M > 4$ define $r_{\text{bal}}(M)$ by

$$r_{\text{bal}}(M) + 1 := \frac{\log(A_M/B_M)}{\log(b_M/a_M)}, \tag{17.1}$$

the unique real order at which the two exact mode bounds $A_{M,r} = A_M a_M^{r+1}$ and $B_{M,r} = B_M b_M^{r+1}$ of (16.4)–(16.6) are equal.

Proposition 17.2 (Asymptotic balance law and window). *As $M \to \infty$ through powers of two,*

$$r_{\mathrm{bal}}(M) = \frac{M \log 2}{\log M}\Big(1 + \mathcal{O}\big(\tfrac{1}{\log M}\big)\Big); \tag{17.2}$$

the orders at which the two modes separately cross 1 are $r_\pm + 1 \sim M\log 2/\log(M/2)$ and $\sim M\log 2/\log(M/4)$, so the window in which both exponential mechanisms are simultaneously tamed has width only

$$r_+(M) - r_-(M) \sim \frac{M(\log 2)^2}{(\log M)^2}. \tag{17.3}$$

Retaining only the leading logarithms of the two modes (equivalently, balancing $M\log 2 + r(\log 2 - \log M)$ against $-M\log 2 + r(\log M - \log 4)$) gives the closed dyadic form

$$r_{\mathrm{bal}}^{\log}(M) = \frac{2M\log 2}{2\log M - 3\log 2} = \frac{M}{\log_2 M - \frac{3}{2}} \overset{M=2^m}{=} \frac{2^{m+1}}{2m-3}, \tag{17.4}$$

with the same asymptotics.

Proof. Insert the Stirling expansions $\log A_M = M\log 2 - \log M - \log(\pi\sqrt{2}) + \mathcal{O}(M^{-1})$, $\log B_M = -(M-2)\log 2 + \frac{1}{2}\log\frac{2}{\pi M} + \mathcal{O}(M^{-1})$, $\log a_M = -\log(M/2) + \log(1 - \frac{1}{2M})$, $\log b_M = \log(M/4) + \log(1 + \frac{1}{M})$ (Appendix B) into (17.1): the numerator is $2M\log 2 + \mathcal{O}(\log M)$ and the denominator $2\log M + \mathcal{O}(1)$. The thresholds solve $A_{M,r} = 1$ and $B_{M,r} = 1$ respectively, and the window width is an elementary expansion of the difference of reciprocal logarithms. ☐

At finite M the exact order (17.1) and the leading-log order (17.4) differ noticeably — by $\mathcal{O}(M/(\log M)^2)$, the same size as the window — and the observed optima sit between them:

M	$r_{\mathrm{bal}}(M)$ exact	$r_{\mathrm{bal}}^{\log}(M)$	error-optimal r (scanned)	$\mathscr{L}$-optimal r	minimum sampled error
16	4.11	6.40	5	5	2.73×10^{-5}
32	7.16	9.14	8	8	7.62×10^{-7}
64	12.42	14.22	14	14	1.78×10^{-6}
128	21.57	23.27	23	23	3.98×10^{-3}
256	37.76	39.38	40	41	9.33×10^{8}

Table 7: Balance predictions against the scanned optima for F. The error-optimal and stability-optimal orders coincide or nearly coincide, and both track the balance laws to within one or two units — while the attained minimum error itself eventually *diverges* (see section 18). The Rvachev scans give the same optimal orders at $M = 16, 32, 64$, although the $[-1,1]$ problem uses no smoothstep: the window location is governed by node geometry and endpoint multiplicity, the target only by how completely the large modes cancel.

Warning 17.3. The balance order minimizes the *worst-case* amplification envelope, not the error. At $r \approx r_{\mathrm{bal}}$ the sampled weighted Lebesgue constant at $M = 64$ is still $\mathscr{L}_{64,14} \approx 2.8 \times 10^3$ — and the entropy modes (16.8) remain exponentially large. The observed 10^{-6} errors therefore rest on highly structured cancellation in the exact Thue–Morse data, and tiny relative perturbations of the samples can destroy them. Function-specific accuracy and operator stability must never be conflated in this problem.

17.2 Lambert-W appearances

Two independent Lambert structures attach to the window.

Budget inversion. Solving the leading law $r \sim M \log 2/\log M$ for the largest comb a fixed jet budget r can serve gives

$$M_{\max}(r) \sim -\frac{r}{\log 2}\, W_{-1}\Big(-\frac{\log 2}{r}\Big), \tag{17.5}$$

with W_{-1} the lower real branch — the same branch that governs the endpoint asymptotics of F itself in the corpus.

Peak migration. For jet ratio $\rho = r/M$, the exponential rate of the weighted central cardinal at position x is

$$\Phi_\rho(x) = \log 2 - \mathsf{H}(x) + \rho \log\big[4x(1-x)\big], \qquad \Phi'_\rho(x) = \log\frac{x}{1-x} + \rho\,\frac{1-2x}{x(1-x)}, \tag{17.6}$$

and in the regime $\rho \downarrow 0$, $r \to \infty$ the boundary saddle solves $x\log(1/x) \sim \rho$, i.e.

$$x_\rho \sim -\frac{\rho}{W_{-1}(-\rho)}. \tag{17.7}$$

Conjecture 17.4 (Lambert peak law)**.** *In that regime the dominant weighted-cardinal peak associated with central nodes is located at* $x_{M,r} = -\dfrac{r/M}{W_{-1}(-r/M)}\big(1+o(1)\big)$*, up to a bounded log-periodic modulation induced by the dyadic structure of the exact data. (The observed migration of the* error *maximizer at* $r=0$*, table 4, scales like* $1/M$*, the* $\rho \to 0$ *endpoint of the same family.)*

17.3 The all-orders correction program

Both balance laws use only first-order Stirling expansions. Every correction to $\log A_M$, $\log B_M$, $\log a_M$, $\log b_M$ is an explicit Bernoulli-polynomial series, and reversion of (17.1) can be organized with complete Bell polynomials, exactly as in the corpus's all-orders Lambert expansions. The result is a computable hierarchy

$$r_{\mathrm{opt}}(M) \sim \frac{M\log 2}{\log M}\Big(1 + \frac{c_1}{\log M} + \frac{c_2 \log\log M}{(\log M)^2} + \cdots\Big), \tag{17.8}$$

whose coefficients for the *actual* optimum should acquire function-specific Thue–Morse corrections. Since the exact order (17.1) is trivial to evaluate numerically, the value of (17.8) is conceptual: it places the interpolation window in the same asymptotic calculus (Stirling, Bernoulli, Bell, Lambert) as the Fabius endpoint itself. Testing the corrected law against exact minimizers at $M = 512, 1024$ is a concrete computational project (section 21).

18 High-precision experiments

18.1 Protocol

All three absorbed interpolation studies follow the same discipline, and their overlapping measurements agree to all displayed digits:

1. every interpolation datum (value or derivative) is an exact rational, produced by the evaluators of section 1.4 and the derivative tower (1.7);
2. interpolants are evaluated by scaled barycentric formulas ((14.1), (15.10)) in `mpmath`, at 70–300 decimal digits, raised until every displayed digit stabilizes;

3. error maxima are *sampled* on complete dyadic grids of stated mesh (2^{-11}–2^{-13}), not certified continuous suprema; symmetry restricts the search to $[0, \frac{1}{2}]$;
4. independent formulations cross-check: ordinary interpolation is computed both directly and as $H_{M,0}$; full-grid Hermite uses confluent divided differences; symmetry, nodal reproduction, endpoint jets, and known small values are verified before each scan.

The scripts, raw CSV data, and figures live under `assets/` in the three interpolation packages.

18.2 Endpoint jets: suppression, window, reversal

Target	M	$r=0$	$r=1$	$r=2$	$r=4$	$r=8$
F	8	9.49×10^{-3}	1.75×10^{-3}	1.50×10^{-3}	1.64×10^{-3}	2.78×10^{-2}
F	16	3.80×10^{-3}	2.68×10^{-4}	1.37×10^{-4}	4.93×10^{-5}	1.44×10^{-4}
F	32	4.99	1.82×10^{-1}	9.75×10^{-3}	1.11×10^{-4}	7.62×10^{-7}
F	64	3.59×10^{7}	6.26×10^{5}	1.68×10^{4}	3.18×10^{1}	4.76×10^{-3}
up	8	2.02×10^{-1}	3.07×10^{-2}	1.96×10^{-2}	1.34×10^{-2}	2.55×10^{-1}
up	16	1.97×10^{-1}	2.67×10^{-2}	4.17×10^{-3}	9.85×10^{-4}	1.06×10^{-2}
up	32	4.50×10^{2}	1.46×10^{1}	6.76×10^{-1}	4.45×10^{-3}	4.08×10^{-5}
up	64	1.05×10^{10}	1.75×10^{8}	4.52×10^{6}	7.72×10^{3}	9.49×10^{-1}

Table 8: Sampled maximum errors at fixed jet orders (up on the symmetric comb of $[-1, 1]$; level-11/-12 check grids). Low-order constraints already buy several orders of magnitude at moderate M.

Scanning all orders at $M = 64$ exposes the sharp U-shape:

r	error for F	error for up	sampled $\mathscr{L}_{64,r}$
0	3.59×10^{7}	1.05×10^{10}	4.40×10^{16}
8	4.76×10^{-3}	9.49×10^{-1}	6.40×10^{6}
10	1.55×10^{-4}	2.73×10^{-2}	2.20×10^{5}
12	8.30×10^{-6}	1.26×10^{-3}	1.29×10^{4}
13	1.94×10^{-6}	2.12×10^{-4}	—
14	1.78×10^{-6}	6.19×10^{-4}	2.83×10^{3}
16	7.79×10^{-5}	1.95×10^{-2}	4.37×10^{4}
18	3.25×10^{-3}	7.31×10^{-1}	—
20	1.72×10^{-1}	3.65×10^{1}	8.46×10^{7}
24	1.13×10^{3}	—	5.49×10^{11}

Table 9: The $M = 64$ jet scan. Both targets are U-shaped with minima at $r = 13$–14, exactly where the sampled weighted Lebesgue constant is also smallest — but that smallest value is still $\approx 2.8 \times 10^3$.

At larger sizes the optimized error stops improving and then diverges:

M	ordinary error	best scanned error (at r)	$\mathscr{L}_{M,r}$ at that r
32	4.99	7.62×10^{-7} (8)	1.30×10^{1}
64	3.59×10^{7}	1.78×10^{-6} (14)	2.83×10^{3}
128	3.84×10^{22}	3.98×10^{-3} (23)	6.19×10^{9}
256	—	9.33×10^{8} (40)	4.28×10^{24}

Table 10: Best endpoint-flat errors across sizes. At $M = 128$ the optimum has already deteriorated to 10^{-3}; at $M = 256$ optimization over all jet orders no longer produces a usable approximation — the first unambiguous sign of the second divergence regime conjectured in theorem 21.2.

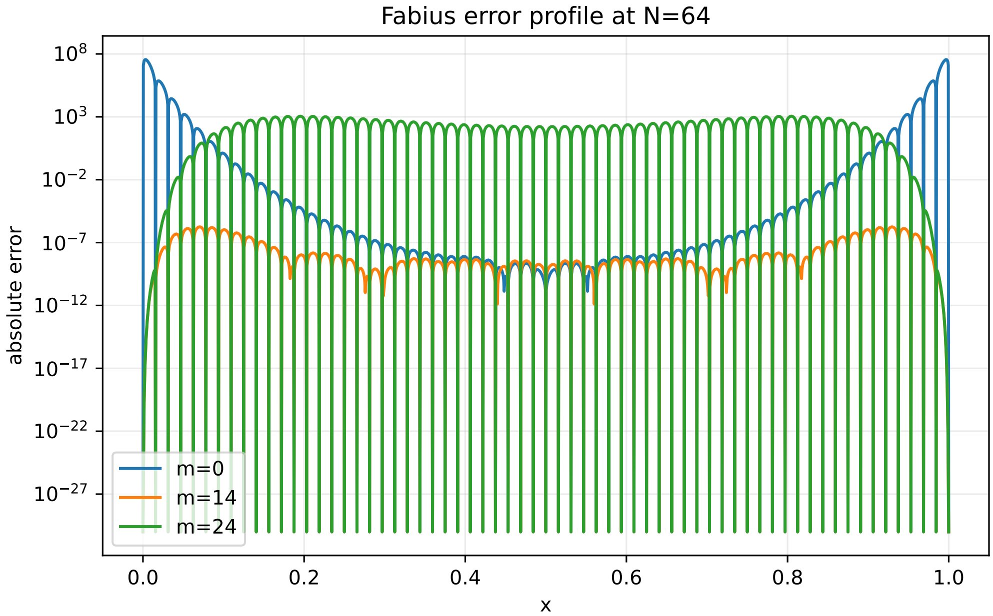

Figure 1: Absolute error profiles at $M = 64$: the ordinary interpolant ($r = 0$) with its giant boundary spike, the tuned order $r = 14$, and the overconstrained $r = 24$ with the reverse instability emerging from the center. (Figure from the absorbed source with N denoting the comb size M.)

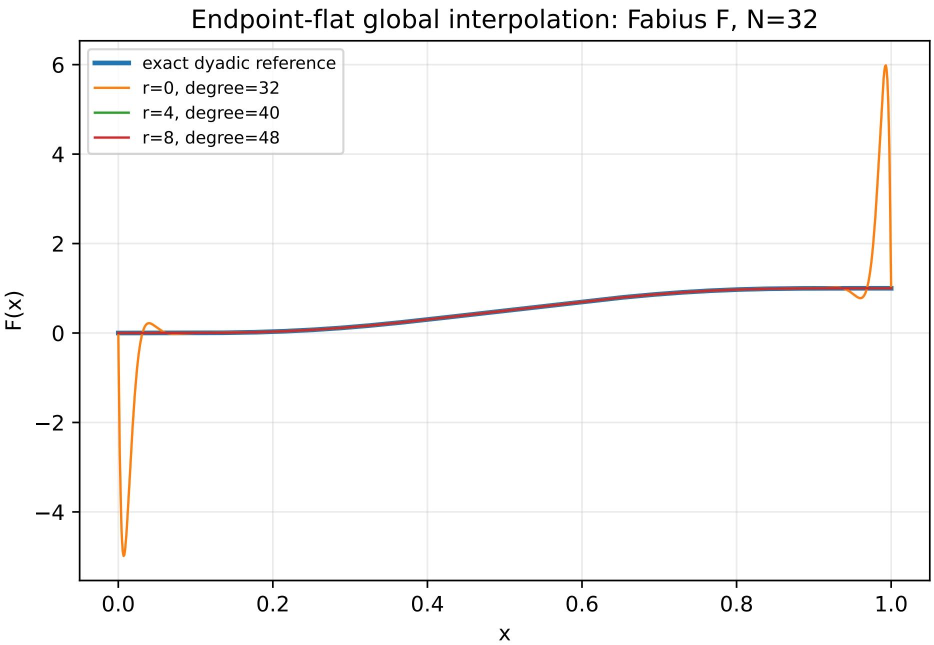

Figure 2: Fabius interpolation on the $M = 32$ comb: ordinary interpolation has entered the boundary-layer regime while moderate endpoint jets recover excellent accuracy — temporarily.

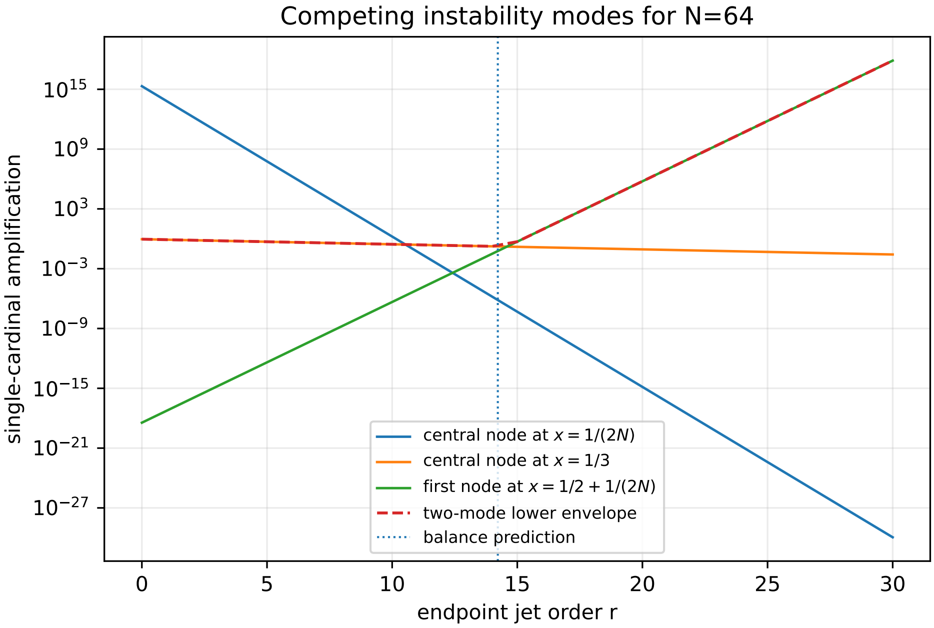

Figure 3: The explicit value-cardinal modes at $M = 64$ as functions of the jet order: raising r suppresses the boundary-to-center and entropy modes but amplifies the first-interior-node mode; their envelope — the mechanism of theorem 16.4 — never becomes small.

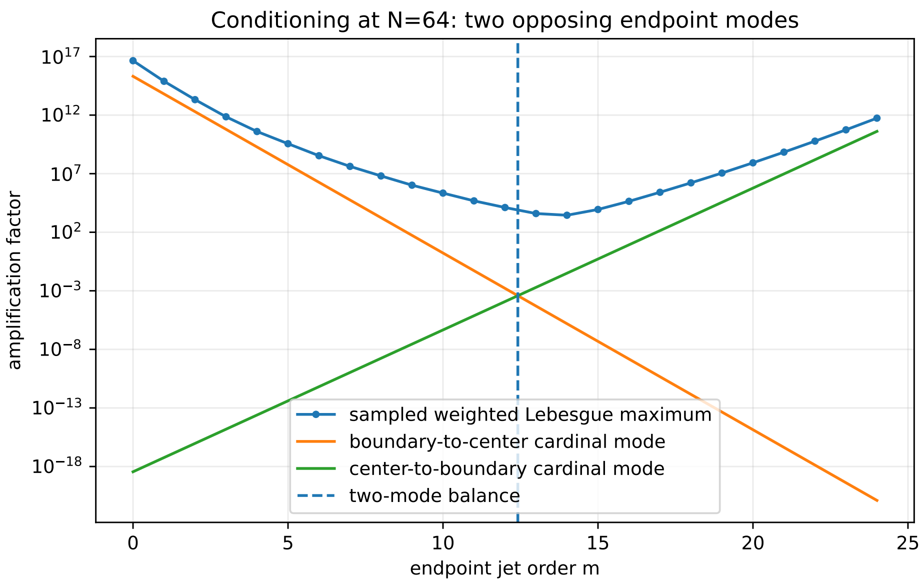

Figure 4: Sampled weighted Lebesgue maximum at $M = 64$ against the two exact mode lower bounds; their crossing predicts the useful finite-M window.

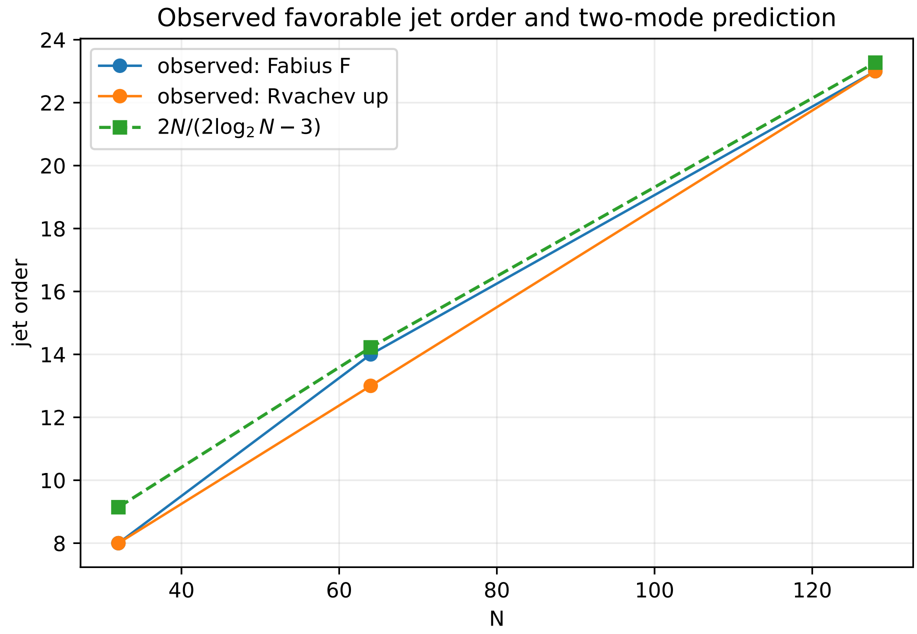

Figure 5: Observed optimal jet orders against the balance law (17.4).

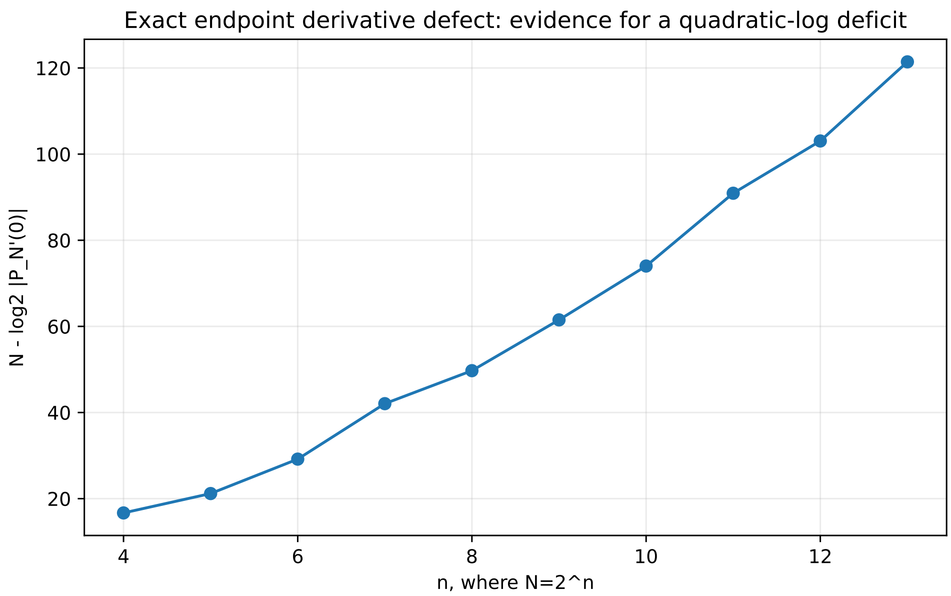

Figure 6: The binary deficit $M - \log_2 |(P_M)'(0)|$ of table 5, consistent with a quadratic in $\log_2 M$ plus oscillation (theorem 14.6).

18.3 Interpretation

1. *Runge behavior occurs for the specific functions*, not merely for the operator: exact data, high precision, and refined check grids leave the explosions unchanged.
2. *Correct endpoint derivatives matter at finite M*: they remove the first spurious jets (14.6) and can improve errors by factors up to $\sim 10^{13}$.
3. *The benefit has a finite window* of width $\asymp M/(\log M)^2$, centered on the balance order; past it, the $W_r(x_j)^{-1}$ amplification takes over, exactly as the mode geometry predicts.
4. *Function-specific cancellation is not stability*: at the optimum the operator amplification is still 10^3-scale at $M = 64$ and grows without bound; by $M = 256$ no jet order helps. The cancellation that produces the 10^{-6} errors is a property of the Thue–Morse structure of the exact samples — understanding it is the arithmetic frontier of section 19.

19 The dyadic hierarchy and arithmetic connections

19.1 Nested corrections and the midpoint surplus

Dyadic combs are nested, so successive interpolants differ by exactly factorable corrections:

$$P_{2M}(x) - P_M(x) = \omega_M(x)\, R_{M-1}(x), \qquad \omega_M(x) = \prod_{j=0}^{M}\Big(x - \frac{j}{M}\Big), \tag{19.1}$$

$$H_{2M,r}(x) - H_{M,r}(x) = \big[x(1-x)\big]^{r+1} \prod_{j=1}^{M-1}\Big(x - \frac{j}{M}\Big)\, R_{M-1,r}(x), \tag{19.2}$$

where in each case the quotient $R_{M-1,\cdot} \in \Pi_{M-1}$ is determined by the M *midpoint surpluses*

$$\sigma_{M,j} = F\left(\frac{2j+1}{2M}\right) - P_M\left(\frac{2j+1}{2M}\right) \qquad (\text{resp. } H_{M,r}), \tag{19.3}$$

by degree counting: both differences vanish at every coarse node (and carry the endpoint multiplicities in the flat case), and the midpoint values fix the quotient. The surpluses are exact rationals, so the hierarchy is a roundoff-free Runge diagnostic: a boundary surplus whose normalized size grows signals the instability of the next level before it is computed, and cells with small surpluses can stop refining — a mathematically transparent adaptive scheme specialized to nested dyadic data.

19.2 Thue–Morse cancellation and the mixed transform

Every Newton coefficient $\Delta^k F(0)$ and every cardinal evaluation is a finite signed sum of exact dyadic values, hence (by theorem 1.2) a finite Thue–Morse expression. Prouhet cancellation annihilates low-degree polynomial trends of complete blocks — the same mechanism as theorem 4.2 — which is the structural reason the function-specific error near r_{bal} can be so much smaller than the operator norm: large cardinals meet blockwise signed moment identities in the data. The cleanest single observable is the endpoint defect (14.7),

$$D_M := (P_M)'(0) = M \sum_{j=1}^{M} (-1)^{j-1} \binom{M}{j} \frac{F(j/M)}{j} = M \int_0^1 \sum_{j=1}^{M} (-1)^{j-1} \binom{M}{j} F\Big(\frac{j}{M}\Big)\, t^{j-1}\, \mathrm{d}t, \tag{19.4}$$

a *mixed transform*: an ordinary (q=1) alternating Pascal row applied to q=$\frac{1}{2}$-structured data. The integral form invites a saddle analysis after inserting the exact evaluator; the entropy concentration of the Pascal row near $j = M/2$, the Prouhet block cancellation, the lower-Lambert endpoint scale of $F(j/M)$ for small j, and a residual log-periodic phase all compete in table 5. A rigorous asymptotic for (19.4) would connect several currently separate parts of the corpus and would settle theorem 14.6.

The contrast between the two node geometries in the corpus is now sharp:

Geometry	Nodes	Natural weights
additive dyadic comb	$j/2^m$	$(-1)^{j-1}\binom{M-2}{j-1}$ (2-adic law (14.5))
geometric/q comb	q^j or -2^j	Gaussian-binomial products

Both are Vandermonde inverses, over different node groups. A mixed transform interpolating across additive location while filtering across dyadic scale — for instance through the nested surplus polynomials (19.2) — is a natural place to look for an additive–multiplicative bridge; the valuation-selective filters of section 12 are the Part-I face of the same idea.

19.3 Fourier, Lambert, and inverse-function connections

Spectral view. A global polynomial has no localized Fourier response: its boundary oscillation is broad spectral leakage. A cellwise operator (below) acts as a compactly supported quasi-interpolant whose multiplier can be compared with the sinc product (1.8) near the origin — a route to sharpened local error constants that respects the compact-support structure of up.

Lambert phases. The corpus's endpoint asymptotics carry a nonconstant periodic correction in a lower-Lambert phase. Three observables here plausibly lock to it: the boundary-layer maximizer (scale $1/M$ at $r = 0$, theorem 17.4 in general), the fluctuation $r_{\text{opt}} - r_{\text{bal}}$, and the log-periodic component of the derivative-defect deficit (theorem 14.6).

Inverse function. F^{-1} has infinite endpoint *slopes*, the opposite pathology to flatness. Direct equispaced interpolation of F^{-1} must fail at least as badly, and the endpoint-flat machinery does not transfer; the natural schemes are parametric interpolation of the exact pairs $(F(j/M), j/M)$, a Lambert-coordinate transformation that removes the known singular asymptotic before interpolating the slowly varying correction, or monotone local rational/Hermite methods. The Part-I comb calculus faces the same issue for $\sum x_k^p F^{-1}(x_k)$ (section 12).

20 Stable reconstruction on the same comb

The negative results concern one global polynomial whose degree grows with the sample count. The same exact samples support provably stable schemes; the global/local contrast is the practical conclusion of Part II.

20.1 Local Hermite interpolation: the cure

For $s \geq 0$ let $\mathcal{H}^{\text{loc}}_{M,s} f$ be the piecewise polynomial of degree $2s+1$ matching $f^{(k)}$, $0 \leq k \leq s$, at both ends of every cell $[j/M, (j+1)/M]$ — all data exact rationals via the derivative tower.

Theorem 20.1 (Local Hermite error, with exact Fabius constants). *For $f \in C^{2s+2}$,*

$$\left\| f - \mathcal{H}^{\text{loc}}_{M,s} f \right\|_\infty \leq \frac{\|f^{(2s+2)}\|_\infty}{(2s+2)!} \left(\frac{1}{4M^2} \right)^{s+1}; \tag{20.1}$$

in particular, inserting (1.7),

$$\boxed{\left\| F - \mathcal{H}^{\text{loc}}_{M,s} F \right\|_\infty \leq \frac{2^{(s+1)(2s+1)}}{(2s+2)!} M^{-2s-2}, \qquad \left\| \mathrm{up} - \mathcal{H}^{\text{loc}}_{M,s}\, \mathrm{up} \right\|_\infty \leq \frac{2^{\binom{2s+3}{2}}}{(2s+2)!} M^{-2s-2},} \tag{20.2}$$

the second bound for M equal cells on $[-1, 1]$. The local value and derivative cardinals have norms bounded independently of M: the scheme is uniformly stable.

Proof. On a cell of length h_0 the two-point Hermite remainder is $f^{(2s+2)}(\xi)\,(x-a)^{s+1}(x-b)^{s+1}/(2s+2)!$, bounded by $(h_0^2/4)^{s+1}$ times the derivative norm. For F, $h_0 = 1/M$ and the factor $4^{-(s+1)}$ lowers the derivative exponent $\binom{2s+3}{2}$ to $(s+1)(2s+1)$; for up on $[-1,1]$, $h_0 = 2/M$ makes $h_0^2/4 = M^{-2}$ with no extra power of two. Rescaling any cell to $[0,1]$ shows the local cardinal basis depends only on s. □

With exact values and exact derivatives ($s = 1$, cubic case) the predicted M^{-4} convergence is observed cleanly:

M	4	8	16	32	64
sampled error	9.67×10^{-4}	5.03×10^{-4}	2.06×10^{-5}	2.29×10^{-6}	1.58×10^{-7}

Table 11: Piecewise-cubic Hermite errors for F (level-12 check grid): steady algebraic convergence on the very samples where the global polynomial explodes.

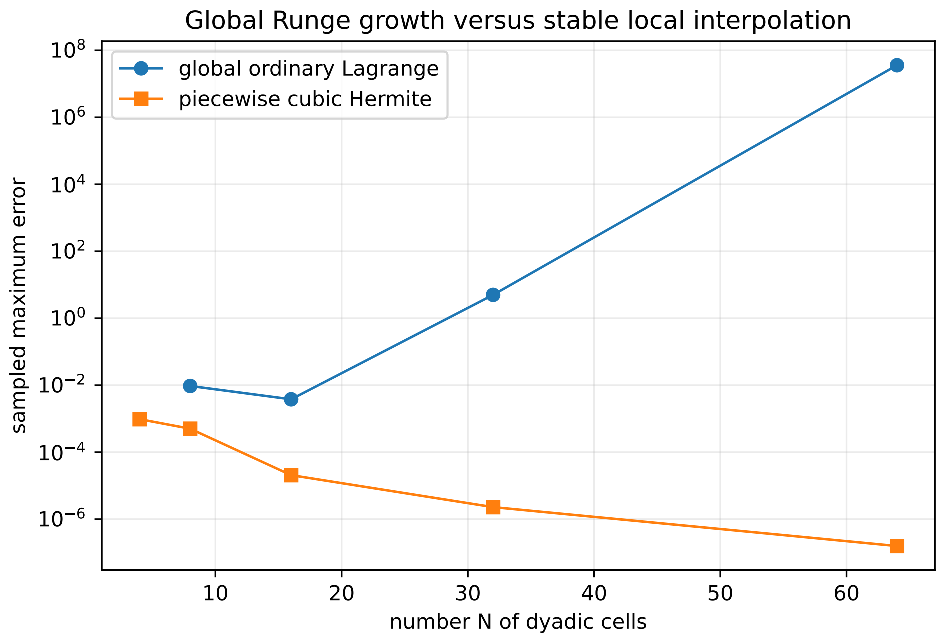

Figure 7: Global ordinary interpolation versus the cellwise cubic method on identical dyadic samples.

20.2 The wider menu

Between the failed global extreme and the local cure lies a hierarchy of methods that keep the exact dyadic data:

- *Oversampled least squares.* Fit degree $d \asymp \sqrt{M}$ to all $M+1$ samples, optionally imposing exact endpoint flatness through the factorization (15.4) with Q fitted rather than interpolated: stable root-exponential accuracy is the best compatible with equispaced data [25, 30].
- *Mock-Chebyshev subset interpolation.* Interpolate only on the comb points nearest to Chebyshev nodes, or constrain a least-squares fit to them; constrained mock-Chebyshev methods extend to Hermite data [31], for which the dyadic derivative tower supplies exact samples.
- *Rational barycentric families.* Floater–Hormann interpolants have logarithmic Lebesgue growth on equispaced nodes [28, 29], and rational Hermite variants accept the exact derivative data; applying them to the deflated quotient g_r preserves endpoint flatness exactly.

- *Mapped and Lambert coordinates.* Interpolate in a coordinate that respects $x \leftrightarrow 1-x$ and opens the endpoint scale — a Lambert coordinate fitted to the known endpoint asymptotics being the natural non-generic choice (cf. Kosloff–Tal-Ezer maps).
- *Atomic/multiresolution bases.* Expand in scaled translates of up itself (the refinement equation makes the dyadic recursion exact), or in Faber–Schauder/spline systems — the representation-theoretic route developed elsewhere in the corpus, with certified local error control.
- *Constrained minimax.* Impose $p(0)=0$, $p(1)=1$, $p^{(q)}(0)=p^{(q)}(1)=0$ and minimize the residual on the comb in a stable basis for g_r.

Practical hierarchy

(1) exact dyadic evaluation for data; (2) local piecewise/atomic reconstruction for stability; (3) global approximation only with degree well below the sample count or after a change of basis/geometry; (4) full-degree global interpolation only for small M or as an exact symbolic identity — never as an asymptotic approximation scheme.

21 Conjectures and frontier problems for Part II

The proved landscape is: operator instability at every jet order (theorems 16.3, 16.4 and 16.6), exact structural identities (theorem 15.1, theorem 16.2, theorem 14.2), and a stable local alternative (theorem 20.1). The open frontier concerns the *specific* Fabius data. Besides theorems 14.5, 14.6, 16.7 and 17.4:

Conjecture 21.1 (Optimal jet order). *Let r_M^F and r_M^{up} minimize the true uniform errors of the endpoint-flat families, and let $r_M^{\mathscr{L}}$ minimize $\mathscr{L}_{M,r}$. All three satisfy*

$$r_M = \frac{M\log 2}{\log M}\big(1+o(1)\big), \qquad \textit{indeed} \qquad r_M = r_{\mathrm{bal}}(M) + \mathcal{O}\bigg(\frac{\log M}{\log\log M}\bigg), \tag{21.1}$$

with a stronger $\mathcal{O}(1)$ version along phase-locked dyadic subsequences. (Scanned data: table 7.)

Conjecture 21.2 (Optimized divergence). $E_M^* := \min_{r\ge 0}\|F-H_{M,r}\|_\infty$ *does not tend to zero; more strongly* $\log E_M^*$ *is eventually linear in M up to $o(M)$, and for the Rvachev version* $\limsup E_M^{*,\mathrm{up}} = +\infty$. *The measured sequence* $7.6\times 10^{-7} \to 1.8\times 10^{-6} \to 4.0\times 10^{-3} \to 9.3\times 10^{8}$ *at* $M = 32, 64, 128, 256$ *(table 10) is the evidence; a proof must show the Thue–Morse cancellation cannot keep pace with the exponential envelope of theorem 16.4.*

Conjecture 21.3 (Boundary-layer profile). *Along a dyadic subsequence there are normalizations A_M with $A_M\big(P_M(\xi/M)-F(\xi/M)\big)\to\mathcal{E}(\xi)$ locally for $\xi>0$, the profile governed jointly by the entropy saddle of the uniform cardinals and the lower-Lambert endpoint asymptotic of F.*

Conjecture 21.4 (Surplus arithmetic). *After clearing the natural common denominator (10.1), the midpoint surplus vector $(\sigma_{M,j})_j$ of (19.3) factors as a finite binomial transform of a Thue–Morse block; its sign changes and its largest boundary entries obey a depth-m recursion. (This would make the nested hierarchy of section 19 an arithmetic object and is the most concrete route to theorem 14.5: showing a boundary surplus cannot stay small at all dyadic levels.)*

Conjecture 21.5 (2-adic stability counterpart). *The valuation law (14.5) admits a renormalized 2-adic interpolation operator on the dyadic data with polynomially bounded norm growth — the archimedean catastrophe and the 2-adic regularity of the same weights being two valuations of one object. Identify the correct Mahler-type basis and compare with the Newton form (14.2).*

Problem 21.6 (Noise threshold)**.** *For tuned $r = r_{\mathrm{bal}}(M)$, quantify the largest relative perturbation of the exact samples that preserves the observed accuracy. The crude bound $\mathscr{L}_{M,r}^{-1}$ is pessimistic; the structured condition number should project noise onto the self-similar residual subspace.*

Research question 21.7 (Sign structure of the differences)**.** *Do the signs or 2-adic valuations of $\Delta^k F(0)$ follow a Thue–Morse or q-binomial law along $M = 2^m$? Such a law would control the cancellations in* (14.2) *and could turn theorem 14.5 into a lower-bound proof.*

Three research routes toward the divergence conjectures, in increasing ambition: (i) insert the exact evaluator into one selected cardinal evaluation and exploit blockwise Thue–Morse cancellation until a dominant term survives; (ii) run a two-variable saddle analysis of $\Delta^k F(0)$ in (k, m), Bernoulli–Bell corrections included; (iii) prove the surplus recursion of theorem 21.4. On the constructive side, the corrected balance expansion (17.8) should be computed and tested against exact minimizers at $M = 512$ and 1024, separately for F and up.

22 Lean formalization program for Part II

The new results separate cleanly into finite algebra, exact gamma identities, and elementary asymptotics — a good match for the repository's formalization style. In dependency order:

1. *Comb combinatorics*: nodes, barycentric weights, the 2-adic law (14.5), symmetry and degree drop (theorem 14.1), Newton form and the Stirling jet transform (14.6).
2. *Endpoint-flat algebra*: the smoothstep (15.1) as a finite polynomial with its jet and symmetry properties; divisibility by W_r; existence, uniqueness, and the error identity of theorem 15.1.
3. *Cardinal identities*: the finite-product forms of theorem 16.1 and the exact modes (16.4)–(16.8) as factorial/binomial identities (gamma notation as corollaries).
4. *Instability*: explicit Stirling bounds for factorials; the two-regime dichotomy of theorem 16.4; the Hermite squares of theorem 16.6.
5. *Local stability*: the two-point Hermite remainder and the derivative-norm specialization (20.2).
6. *Exact certificates*: kernel-checked rational error values at small M (as statements about finite grids, not continuum suprema), and the exact endpoint defects of table 5.

A natural module decomposition:

Module	Contents
`DyadicCombLagrange`	nodes, weights, symmetry, Newton form, 2-adic law
`EndpointFlatHermite`	smoothstep, factorization, uniqueness, error identity
`WeightedLebesgue`	perturbation operator, cardinal functions
`DyadicCombModes`	exact A/B/entropy-mode identities
`DyadicCombInstability`	fixed-r and all-r exponential lower bounds
`FabiusInterpolationDefect`	exact endpoint derivative transform
`LocalHermiteStability`	cellwise remainder and Fabius constants

This would extend the corpus's geometric-node Lagrange formalization to the additive comb without duplicating it; the Part-I roadmap (section 12) shares the first module.

A Gamma-product calculations

A.1 Half-cell central cardinal (mode A)

In the scaled variable $y = Mx$ with interior nodes $1, \dots, M-1$ and $j = M/2$, at $y = \frac{1}{2}$: $\prod_{k=1}^{M-1} \left|k - \frac{1}{2}\right| = \Gamma(M - \frac{1}{2})/\Gamma(\frac{1}{2})$; deleting the j factor divides by $\left|j - \frac{1}{2}\right| = \frac{M}{2} - \frac{1}{2}$, and the denominator of the cardinal is $(j-1)!\,(M-1-j)! = \Gamma(M/2)^2$, giving A_M in (16.4). The weight ratio is $\left[\frac{1}{2M}(1-\frac{1}{2M})\right]/\frac{1}{4} = (2M-1)/M^2 = a_M$. Stirling gives $A_M \sim 2^M/(\pi\sqrt{2}\,M)$.

A.2 Near-central boundary-node cardinal (mode B)

For $j=1$ at $y=(M+1)/2$ with $M=2K$: $\left|\prod_{k=2}^{M-1}(y-k)\right| = \left[\Gamma(K-\frac{1}{2})/\Gamma(\frac{1}{2})\right]^2$ (the products on the two sides of the half-integer are equal), and the denominator is $(M-2)! = \Gamma(M-1)$, giving B_M. The weight ratio is $\left[(\frac{1}{2}+\frac{1}{2M})(\frac{1}{2}-\frac{1}{2M})\right]/\left[\frac{1}{M}(1-\frac{1}{M})\right] = (M+1)/4 = b_M$. The duplication formula turns B_M into $\dfrac{2^{2-M}}{\sqrt{\pi}}\,\dfrac{\Gamma((M-1)/2)}{\Gamma(M/2)} \sim 2^{2-M}\sqrt{2/(\pi M)}$; equivalently $B_M = \binom{M-2}{M/2-1}/16^{M/2-1}$.

A.3 The one-third mode

At $y=M/3$ (dyadic M is never divisible by 3): $\prod_{k=1}^{M-1}(M/3-k) = \Gamma(M/3)/\Gamma(1-2M/3)$, and reflection gives $1/\left|\Gamma(1-2M/3)\right| = \Gamma(2M/3)\left|\sin(2\pi M/3)\right|/\pi = \frac{\sqrt{3}}{2\pi}\Gamma(2M/3)$. Removing the central factor $|M/3-M/2| = M/6$ and dividing by $\Gamma(M/2)^2$ gives (16.8); the exponential part of the gamma ratio is $\exp\{M[\frac{1}{3}\log\frac{1}{3}+\frac{2}{3}\log\frac{2}{3}-\log\frac{1}{2}]\} = \mathrm{e}^{M(\log 2-\mathsf{H}(1/3))}$.

A.4 Full-comb central cardinal

For the full node set $0,\dots,M$ at $y=\frac{1}{2}$, $j=M/2$: $\prod_{k=0}^{M}\left|\frac{1}{2}-k\right| = \frac{1}{2}\Gamma(M+\frac{1}{2})/\Gamma(\frac{1}{2})$; deleting the central factor divides by $\left|\frac{1}{2}-\frac{M}{2}\right| = \frac{M-1}{2}$, and the denominator is $(M/2)!\,(M/2)! = \Gamma(\frac{M}{2}+1)^2$, giving (16.10); Stirling's formula then yields $\left|\ell_{M/2}(\frac{1}{2M})\right| \sim \sqrt{2}\,2^M/(\pi M^2)$, whose square is (16.11).

B Mode asymptotics

Stirling's expansion in logarithmic form,

$$\log\Gamma(z) = \left(z-\tfrac{1}{2}\right)\log z - z + \tfrac{1}{2}\log(2\pi) + \sum_{r=1}^{R-1}\frac{B_{2r}}{2r(2r-1)\,z^{2r-1}} + \mathcal{O}\big(z^{-2R+1}\big), \tag{B.1}$$

applied to (16.4)–(16.6) yields

$$\log A_M = M\log 2 - \log M - \log(\pi\sqrt{2}) + \mathcal{O}(M^{-1}), \tag{B.2}$$

$$\log B_M = -(M-2)\log 2 + \tfrac{1}{2}\log\frac{2}{\pi M} + \mathcal{O}(M^{-1}), \tag{B.3}$$

$$\log a_M = -\log(M/2) + \log\big(1-\tfrac{1}{2M}\big), \qquad \log b_M = \log(M/4) + \log\big(1+\tfrac{1}{M}\big), \tag{B.4}$$

which prove theorem 17.2. Retaining further Bernoulli terms and reverting with complete Bell polynomials yields the all-orders program (17.8).

C Row reciprocity

With $A_m(z) = \sum_{k=0}^{M-1} F(k/M)z^k$ and $F((M-k)/M) = 1-F(k/M)$,

$$z^M A_m(1/z) = \sum_{k=1}^{M} F\big((M-k)/M\big)z^k = \sum_{k=1}^{M}\big(1-F(k/M)\big)z^k = \frac{z(1-z^M)}{1-z} - A_m(z) - z^M, \tag{C.1}$$

which is (3.9). This exact reciprocity forces the palindromic structure of the low-level row numerators (3.6)–(3.8).

D Core algorithms

Listing 1: Streaming the exact dyadic Fabius row (theorem 1.4); d holds the exact Fabius-law moments of (1.20).

```
def fabius_row_exact(m):
    M = 2**m
    d = fabius_law_moments(m)          # Fractions d_0..d_m
    S = [0]*(m+1)                      # signed prefix power sums
    scale = Fraction(1, 2**(m*(m-1)//2) * factorial(m))
    row = []
    for a in range(M+1):
        row.append(scale * sum(comb(m,j)*d[j]*S[m-j]
                               for j in range(m+1)))
        if a == M: break
        eps = -1 if bin(a).count('1') % 2 else 1
        S = [S[0]+eps] + [sum(comb(k,q)*S[q] for q in range(k+1))
                          for k in range(1, m+1)]
    return row
```

Listing 2: Endpoint-flat interpolation in barycentric form (theorem 15.1).

```
S = lambda r,x: (factorial(2*r+1)//(factorial(r)**2)
                 * sum((-1)**k*comb(r,k)*x**(r+k+1)/(r+k+1)
                       for k in range(r+1)))
W = lambda r,x: (x*(1-x))**(r+1)
g = [(F(x_j)-S(r,x_j))/W(r,x_j) for x_j in interior_nodes]
w = [(-1)**(j-1)*comb(M-2,j-1) for j in range(1,M)]
H = S(r,x) + W(r,x)*barycentric(x, interior_nodes, g, w)
```

The absorbed packages add the sparse-difference row generation ($\mathcal{O}(mM)$), confluent divided differences for full-grid Hermite, weighted-Lebesgue scans, and all figure/table generation; entry points and reproduction commands are in each package's README under `assets/`.

E Provenance of the absorbed sources

This volume absorbs the following six drafts (all dated 28 August 2026; SHA-256 of each main `.tex` source at absorption). The supporting material of each — experiment scripts, CSV/data files, figures, READMEs, and the sources' own SHA manifests — is preserved under `assets/<directory>/`; the draft directories themselves are deleted, git history being the archive.

Source (directory; SHA-256 prefix)	Role in this volume
Dyadic-Comb Sums for the Fabius–Rvachev System (`Fabius_Dyadic_Comb_Sums_Report_Package`; `4b6b0384df3f...`)	Row OGF and pole collapse; Bernoulli–Thue–Morse and Bell–Prouhet closed forms; the χ-based collapse proof; odd-level defect series; Hurwitz formula; Stirling reduction of iterated sums; fractional-order semigroup; generated exact tables (tables 1 and 2).
Dyadic-Comb Moments of the Fabius and Rvachev Functions (`fabius-dyadic-comb-sums-report`; `b815be3b441e...`)	Moment EGF; probabilistic Irwin–Hall proof of the dyadic convolution; sparse $\mathcal{O}(mM)$ algorithm; one-sided transform; spectral Dirichlet values $D(2), D(4)$ and their rationality; midpoint identity; Newton fractional series; Richardson filtering; tensor cubature.
Dyadic-Comb Quadrature for the Fabius and Rvachev Functions (`fabius_dyadic_comb_report_final`; `3379d821b764...`)	Shifted exactness for arbitrary real phase; 2-adic alias hierarchy; Bell–Bernoulli jet calculus; parity threshold $\tau(p)$ and the one-formula closed form; absolute integer powers; cutoff-Mellin lemma, universal zeta correction, shifted Hurwitz corrections, negative-power renormalization; iterated closed forms; base-b extension; phase-superconvergence experiments.
Dyadic-Comb Interpolation of the Fabius and Rvachev Functions (`fabius_dyadic_interpolation_report`; `970d261c4cb7...`)	Streaming evaluator; symmetry compression; smoothstep factorization; half-cell and one-third modes; fixed-r theorem; leading-log balance law; all-node Hermite theorem ($4^M/M^4$); nested corrections and midpoint surplus; local Hermite stability theorem with exact constants; gamma appendix.
Dyadic-Comb Interpolation of the Fabius and Rvachev Functions (independent write-up, same title; `fabius_interpolation_report`; `799ff356366...`)	Bit-recursion evaluator; degree drop; exact finite-M balance order, window width, and thresholds; entropy-mode instability proof; measured weighted Lebesgue constants; exact endpoint derivative defects to $M = 8192$ and the quadratic-log conjecture; mixed $q{=}1/q{=}\frac{1}{2}$ transform; Lambert budget inversion; alternatives survey; Lean module table.
Global Polynomial Interpolation of the Fabius and Rvachev Functions on Dyadic Combs (`Fabius_Rvachev_Dyadic_Interpolation_Report`; `f4d8c1c280ad...`)	2-adic weight valuation law; exact error identity form of the factorization; even reduction of the full Rvachev comb to quadratic nodes; general confluent Hermite data and tables; $M = 256$ optimized-divergence data; ordinary Lebesgue-constant measurements; Lambert peak-migration saddle; mock-Chebyshev/rational-Hermite survey and references.

What was deduplicated. All six sources restate the corpus background of section 1; the three sums sources prove the same collapse theorem (three routes; the χ-aliasing proof is presented here, the operator-identity and one-sided-transform routes surviving as (5.3) and remarks) and the same first-defect series in three normalizations; the three interpolation sources prove the same factorization theorem, the same fixed-order instability, and three variants of the all-order instability (the entropy-mode proof is presented here; the $x = \frac{1}{3}$ and $a = \frac{1}{10}$ variants survive as the exact identity (16.8) and the choice of a in the proof). Their numerical tables overlap heavily and agree wherever they overlap; each retained table cites its generating package. The two new exact values $D(6), D(8)$ of (6.6) were computed for this volume as described in section 6. Beyond notation, no source's mathematical content was discarded.

Each source carried its own audit of the repository corpus against duplication (pinned snapshots from `fa53516a...` to `3c7f2912...`); their unanimous conclusion — no prior additive-comb Runge study, no endpoint-flat factorization, no unweighted-comb collapse theorem in the audited corpus — is inherited here, with the same caveat that this is a corpus-relative and targeted-literature novelty standard, not a claim of universal priority.

References

[1] J. Fabius, A probabilistic example of a nowhere analytic C^∞-function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174. DOI: 10.1007/BF00536652.

[2] B. Jessen and A. Wintner, Distribution functions and the Riemann zeta function, *Transactions of the American Mathematical Society* **38** (1935), 48–88.

[3] V. L. Rvachev and V. A. Rvachev, A certain finite function, *Dopovidi Akademii Nauk Ukrains'koi RSR, Series A* (1971), 705–707, 764.

[4] V. A. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45** (1990), no. 1, 87–120.

[5] J. Arias de Reyna, Definición y estudio de una función indefinidamente diferenciable de soporte compacto, *Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales de Madrid* **76** (1982), no. 1, 21–38; English translation arXiv:1702.05442 (2017).

[6] J. Arias de Reyna, Arithmetic of the Fabius function, *INTEGERS* **18** (2018), Article A51; arXiv:1702.06487, version 3.

[7] J. K. Haugland, Evaluating the Fabius function, arXiv:1609.07999 (2016).

[8] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.

[9] L. Comtet, *Advanced Combinatorics*, Reidel, 1974.

[10] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics*, 2nd ed., Addison–Wesley, 1994.

[11] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, https://dlmf.nist.gov/; chapters on Bernoulli polynomials, Hurwitz zeta, and asymptotic expansions.

[12] G. Strang and G. Fix, A Fourier analysis of the finite element variational method, in *Constructive Aspects of Functional Analysis*, C.I.M.E., 1973, pp. 793–840.

[13] C. de Boor and A. Ron, Fourier analysis of the approximation power of principal shift-invariant spaces, *Constructive Approximation* **8** (1992), 427–462.

[14] L. N. Trefethen and J. A. C. Weideman, The exponentially convergent trapezoidal rule, *SIAM Review* **56** (2014), no. 3, 385–458.

[15] I. Navot, An extension of the Euler–Maclaurin summation formula to functions with a branch singularity, *Journal of Mathematics and Physics* **40** (1961), 271–276.

[16] N. Berline and M. Vergne, Local asymptotic Euler–Maclaurin expansion for Riemann sums over a semi-rational polyhedron, arXiv:1502.01671 (2015).

[17] A. A. Buchheit and T. Keßler, Singular Euler–Maclaurin expansion, arXiv:2003.12422 (2020).

[18] C. Runge, Über empirische Funktionen und die Interpolation zwischen äquidistanten Ordinaten, *Zeitschrift für Mathematik und Physik* **46** (1901), 224–243.

[19] T. M. Mills and S. J. Smith, The Lebesgue constant for Lagrange interpolation on equidistant nodes, *Numerische Mathematik* **61** (1992), 111–115.

[20] L. Brutman, Lebesgue functions for polynomial interpolation — a survey, *Annals of Numerical Mathematics* **4** (1997), 111–127.

[21] S. J. Smith, Lebesgue constants in polynomial interpolation, *Annales Mathematicae et Informaticae* **33** (2006), 109–123.

[22] J.-P. Berrut and L. N. Trefethen, Barycentric Lagrange interpolation, *SIAM Review* **46** (2004), no. 3, 501–517.

[23] N. J. Higham, The numerical stability of barycentric Lagrange interpolation, *IMA Journal of Numerical Analysis* **24** (2004), 547–556.

[24] D. Coppersmith and T. J. Rivlin, The growth of polynomials bounded at equally spaced points, *SIAM Journal on Mathematical Analysis* **23** (1992), no. 4, 970–983.

[25] R. B. Platte, L. N. Trefethen, and A. B. J. Kuijlaars, Impossibility of fast stable approximation of analytic functions from equispaced samples, *SIAM Review* **53** (2011), no. 2, 308–318.

[26] L. N. Trefethen, *Approximation Theory and Approximation Practice*, SIAM, Philadelphia, 2013.

[27] C. Manni, Lebesgue constants for Hermite and Fejér interpolation on equidistant nodes, *Calcolo* **30** (1993), 93–102.

[28] M. S. Floater and K. Hormann, Barycentric rational interpolation with no poles and high rates of approximation, *Numerische Mathematik* **107** (2007), 315–331.

[29] E. Cirillo and K. Hormann, On the Lebesgue constant of barycentric rational Hermite interpolants at equidistant nodes, *Journal of Computational and Applied Mathematics* **349** (2019), 143–155.

[30] B. Adcock and A. Shadrin, Fast and stable approximation of analytic functions from equispaced samples via polynomial frames, *Constructive Approximation* **57** (2023), 257–294.

[31] F. Dell'Accio, F. Marcellán, and F. Nudo, Constrained mock-Chebyshev least squares approximation for Hermite interpolation, *Numerical Algorithms* (2025), DOI: `10.1007/s11075-025-02051-7`.

[32] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359.

[33] P. Flajolet, X. Gourdon, and P. Dumas, Mellin transforms and asymptotics: harmonic sums, *Theoretical Computer Science* **144** (1995), 3–58.

[34] V. Reshetnikov, *The Fabius Function and Rvachev's Up-Function*, primary exposition in the `ProveIt` repository, `Analysis/FabiusFunction/docs`, snapshot 28 August 2026.

[35] V. Reshetnikov, *Non-Formalized Research Frontiers for the Fabius Function*, `ProveIt` repository, `Analysis/FabiusFunction/docs`, snapshot 28 August 2026.